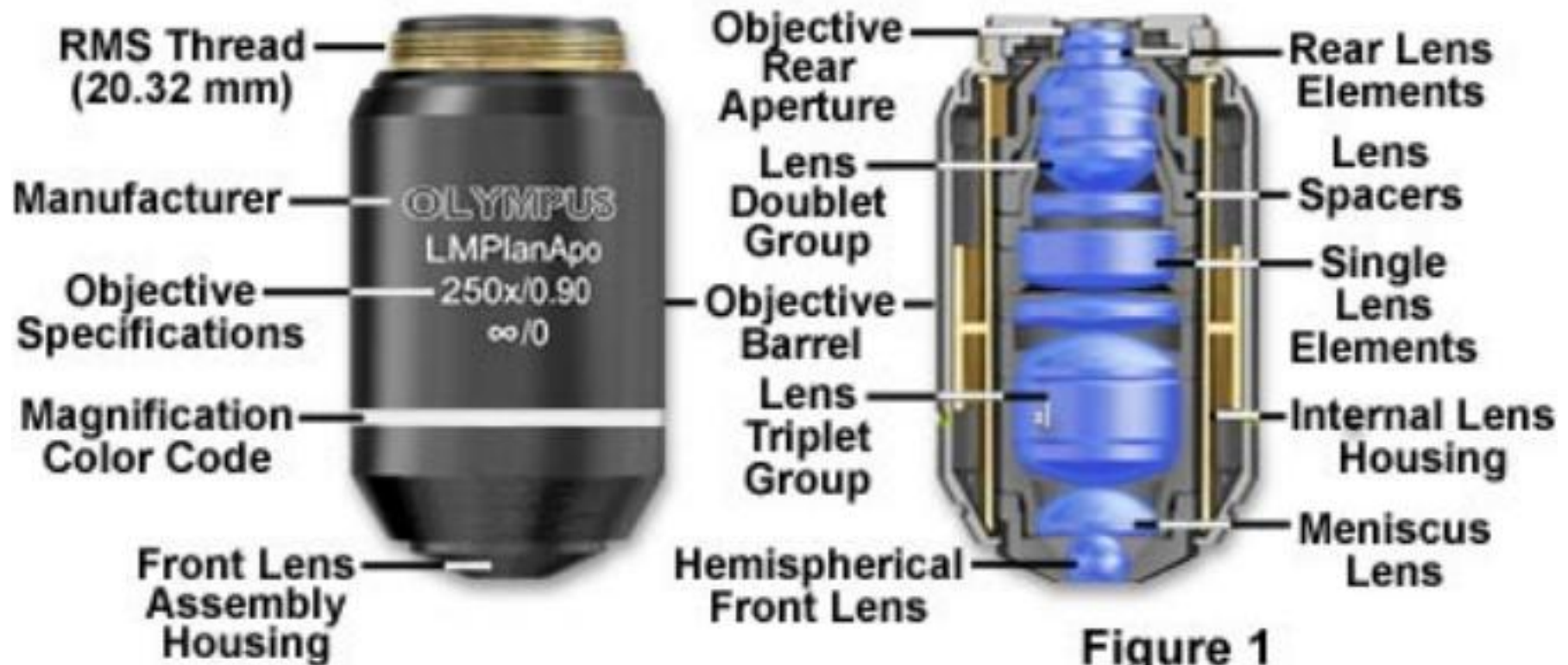


第六章 显微光学

1. 物镜的强聚焦及衍射极限
2. 物镜成像的矢量波动理论
3. 扫描共焦显微镜的成像理论
4. 超分辨成像技术简介

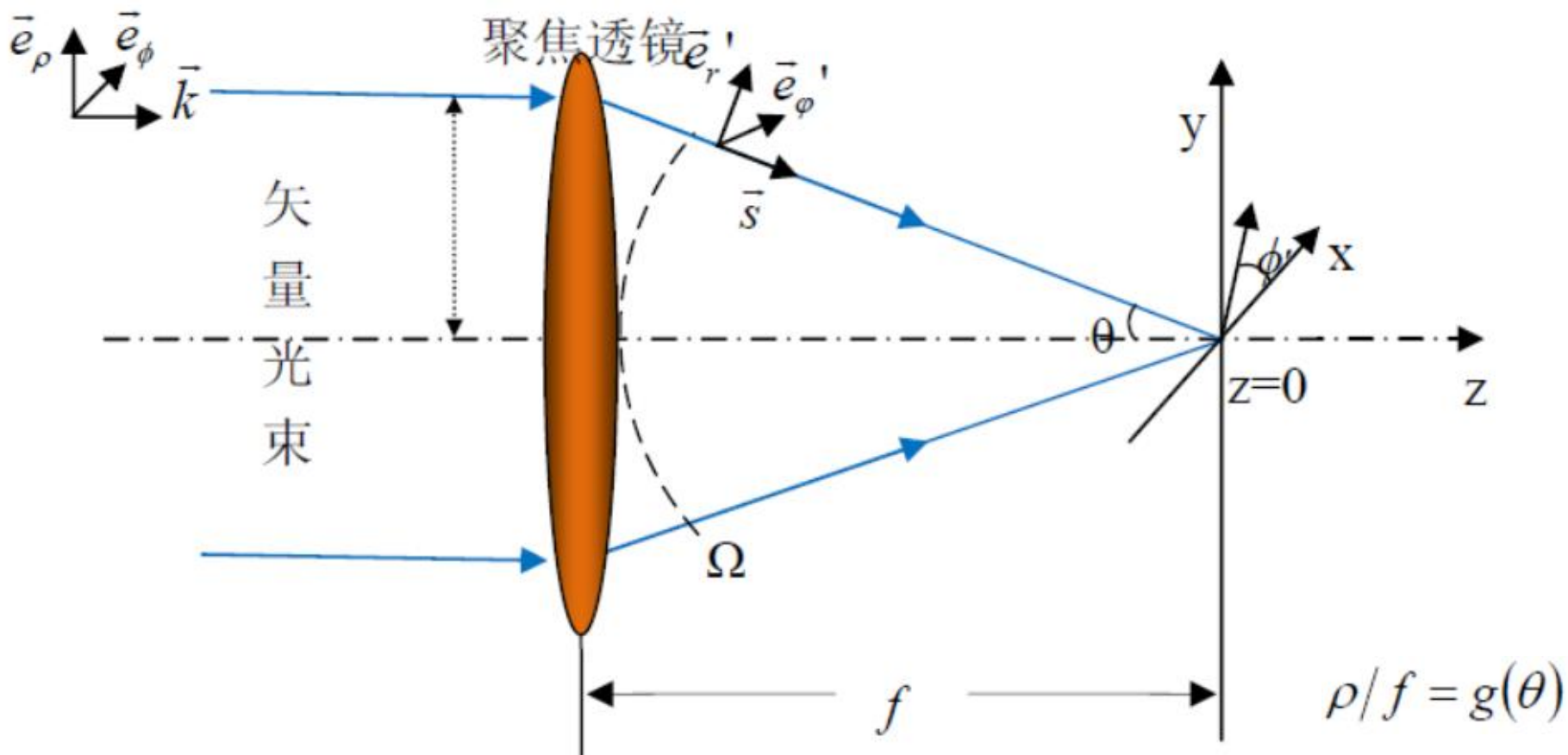
6.1. 物镜的强聚焦及衍射极限

LWD Plan Infinity-Corrected Apochromat Objective



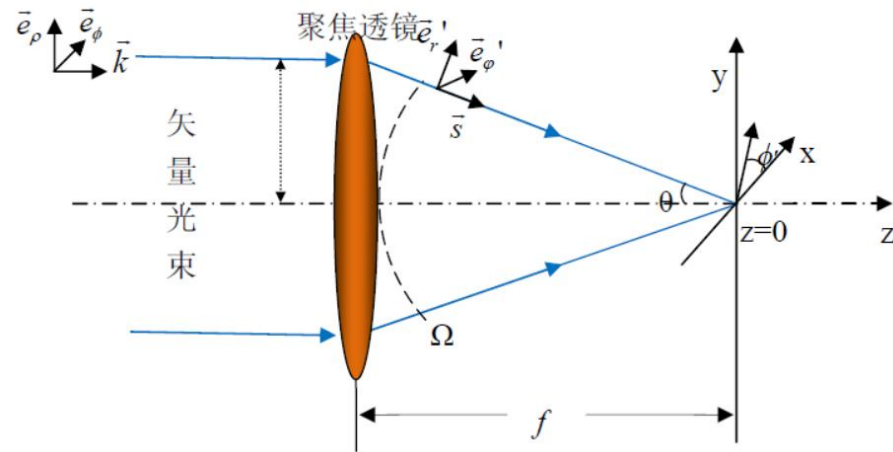
6.1. 物镜的强聚焦及衍射极限

1) 矢量光束的聚焦:



$$\mathbf{E}_i = l_0 P(\rho) [\xi(\rho, \phi) \vec{e}_\rho + \eta(\rho, \phi) \vec{e}_\phi], \quad \xi(\rho, \phi)^2 + \eta(\rho, \phi)^2 = 1$$

6.1. 物镜的强聚焦及衍射极限



前提条件:

1) 物镜是无球差的, 或者, 其球差可以忽略。则, 被物镜会聚后, 在入瞳处, 每个位置上的光都向焦点传播。

2) 如果以焦点为中心、 f 为半径, 画一个球面, 那么球面上每一点处的光都来自于入射光, 且‘一一对映’。即球面上每一点电场 $U(\theta)$ 均可以由入射光 $U_0(\rho)$ 得出。

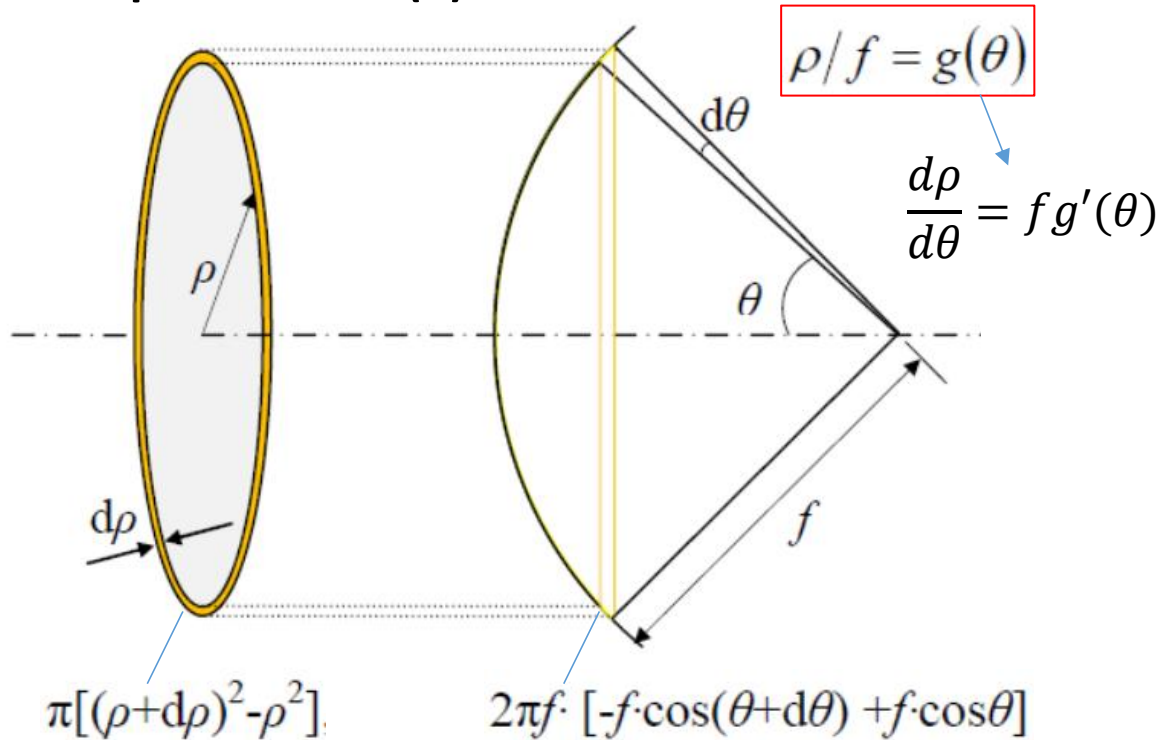
光聚焦时的处理步骤:

- 1) 先求出 $U(\theta)$ 。
- 2) 以 $U(\theta)$ 为波前的会聚球面波, 利用**矢量角谱**方法, 计算其在焦点附近的光场。

6.1. 物镜的强聚焦及衍射极限

Step 1: 先求 $U(\theta)$

1) 复振幅



正弦条件: $g(\theta) = \sin\theta$

Lagrange条件: $g(\theta) = \theta$

Herschel条件: $g(\theta) = 2\sin(\theta/2)$

Helmholtz条件: $g(\theta) = \tan\theta$

满足能量守恒:

$$[I_0 P(\rho)]^2 \cdot \pi[(\rho+d\rho)^2 - \rho^2] = [I_0 P(\theta)]^2 \cdot 2\pi f \cdot [-f \cdot \cos(\theta+d\theta) + f \cdot \cos\theta]$$

$$(d\rho)^2 \approx 0, \sin(d\theta) \approx d\theta, \cos(d\theta) \approx 1$$

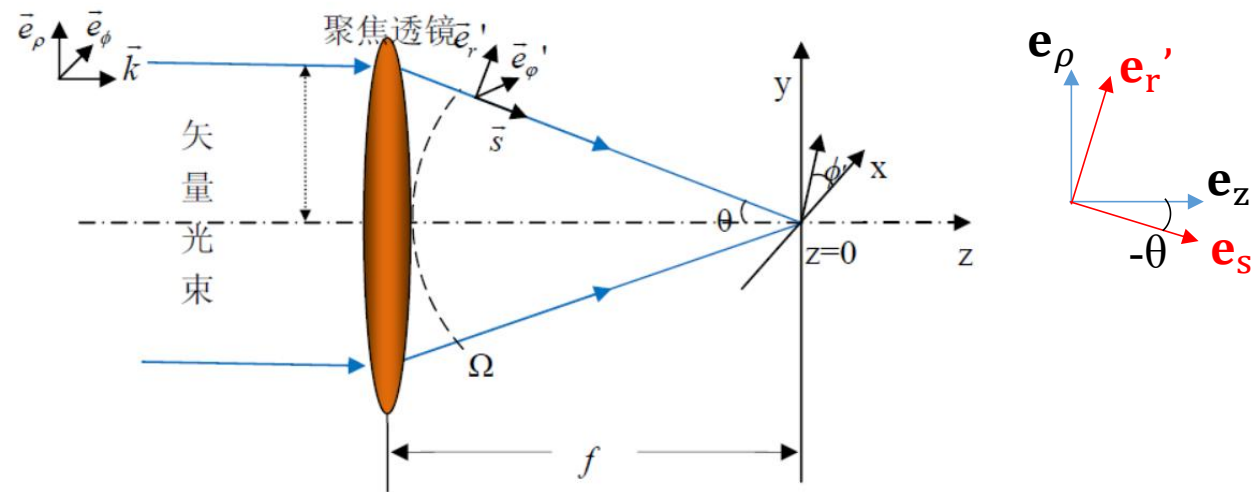
$$P(\rho)^2 2\pi \rho d\rho = P(\theta)^2 \cdot 2\pi f \cdot f \sin\theta d\theta$$

$$P(\theta) = P(\rho) \sqrt{g(\theta)g'(\theta)/\sin\theta} = P(fg(\theta)) \sqrt{g(\theta)g'(\theta)/\sin\theta}$$

6.1. 物镜的强聚焦及衍射极限

Step 1: 先求 $U(\theta)$

2) 偏振



$$\vec{e}_r' = +\cos\theta(\cos\varphi\vec{e}_x + \sin\varphi\vec{e}_y) - \sin\theta\vec{e}_z,$$

$$\vec{e}_\phi' = \vec{e}_\phi = -\sin\varphi\vec{e}_x + \cos\varphi\vec{e}_y,$$

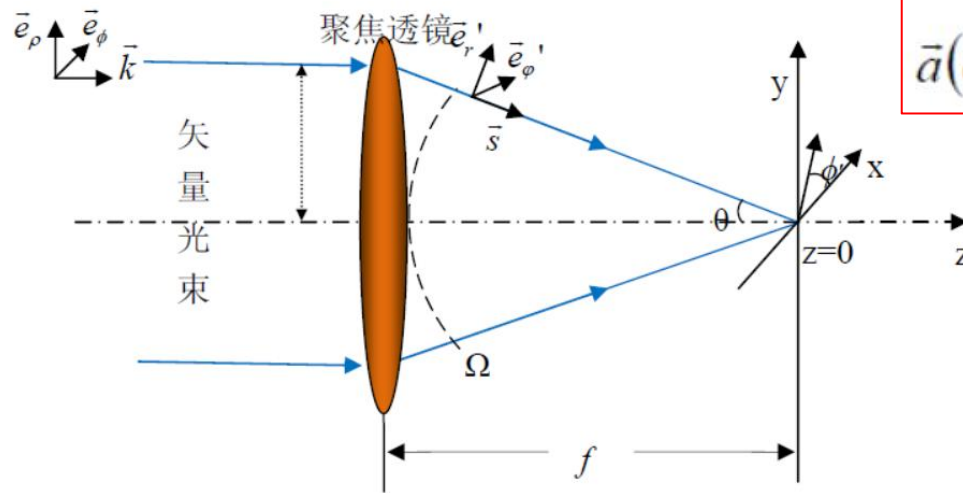
$$\vec{s} = \sin\theta(\cos\varphi\vec{e}_x + \sin\varphi\vec{e}_y) + \cos\theta\vec{e}_z,$$

光线偏折后，偏振态不变

$$\bar{a}(\theta, \varphi) = l_0 f P(\theta) [\xi(fg(\theta), \varphi) \vec{e}_r' + \eta(fg(\theta), \varphi) \vec{e}_\phi']$$

6.1. 物镜的强聚焦及衍射极限

Step 2: 利用矢量角谱处理矢量光的衍射问题



$$\bar{a}(\theta, \varphi) = l_0 f P(\theta) [\xi(fg(\theta), \varphi) \bar{\mathbf{e}}_r' + \eta(fg(\theta), \varphi) \bar{\mathbf{e}}_\varphi']$$

$$\bar{\mathbf{s}} = \sin \theta (\cos \varphi \bar{\mathbf{e}}_x + \sin \varphi \bar{\mathbf{e}}_y) + \cos \theta \bar{\mathbf{e}}_z$$

$$\mathbf{r} = x \bar{\mathbf{e}}_x + y \bar{\mathbf{e}}_y + z \bar{\mathbf{e}}_z$$

$$\bar{\mathbf{r}} = r \cos \phi' \bar{\mathbf{e}}_x + r \sin \phi' \bar{\mathbf{e}}_y + z \bar{\mathbf{e}}_z$$

$$\bar{\mathbf{s}} \cdot \bar{\mathbf{r}} = z \cos \theta + r \sin \theta \cos(\varphi - \phi')$$

矢量角谱方法

$$\mathbf{e}(P) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{e}^{ik(s_x x + s_y y)} [\mathbf{U}(s_x, s_y) \mathbf{e}^{iks_z z}] ds_x ds_y$$

会聚光的矢量角谱方法

$$\begin{aligned} \mathbf{e}(P) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{e}^{ik(s_x x + s_y y)} [\mathbf{U}(s_x, s_y) \mathbf{e}^{iks_z z}] \frac{1}{s_z} ds_x ds_y \\ &= -\frac{ik}{2\pi} \iint_{\Omega} \mathbf{A}(\mathbf{s}) \mathbf{e}^{ik\mathbf{s} \cdot \mathbf{r}} d\Omega \end{aligned}$$

$$\frac{ds_x ds_y}{s_z} = d\Omega$$

Debye积分

6.1. 物镜的强聚焦及衍射极限

$$\bar{a}(\theta, \varphi) = l_0 f P(\theta) [\xi(fg(\theta), \varphi) \bar{\mathbf{e}}_r' + \eta(fg(\theta), \varphi) \bar{\mathbf{e}}_\varphi']$$

$$\bar{\mathbf{e}}_r' = +\cos\theta(\cos\varphi\bar{\mathbf{e}}_x + \sin\varphi\bar{\mathbf{e}}_y) - \sin\theta\bar{\mathbf{e}}_z$$

$$\bar{\mathbf{e}}_\varphi' = \bar{\mathbf{e}}_\phi = -\sin\varphi\bar{\mathbf{e}}_x + \cos\varphi\bar{\mathbf{e}}_y,$$

$$\mathbf{E}(r, \phi', z) = \frac{-ik}{2\pi} \int_0^{\theta_{\max}} d\theta \int_0^{2\pi} l_0 f P(\theta) [\xi(fg(\theta), \varphi) \bar{\mathbf{e}}_r' + \eta(fg(\theta), \varphi) \bar{\mathbf{e}}_\varphi']$$

$$e^{ik[z \cos\theta + r \sin\theta \cos(\varphi - \phi')]} \sin\theta d\varphi$$

$$= \frac{-il_0 f}{\lambda} \int_0^{\theta_{\max}} d\theta \int_0^{2\pi} P(\theta) \left[\xi(fg(\theta), \varphi) \begin{pmatrix} \cos\theta \cos\varphi \bar{\mathbf{e}}_x \\ \cos\theta \sin\varphi \bar{\mathbf{e}}_y \\ \sin\theta \bar{\mathbf{e}}_z \end{pmatrix} + \eta(fg(\theta), \varphi) \begin{pmatrix} -\sin\varphi \bar{\mathbf{e}}_x \\ \cos\varphi \bar{\mathbf{e}}_y \\ 0 \bar{\mathbf{e}}_z \end{pmatrix} \right]$$

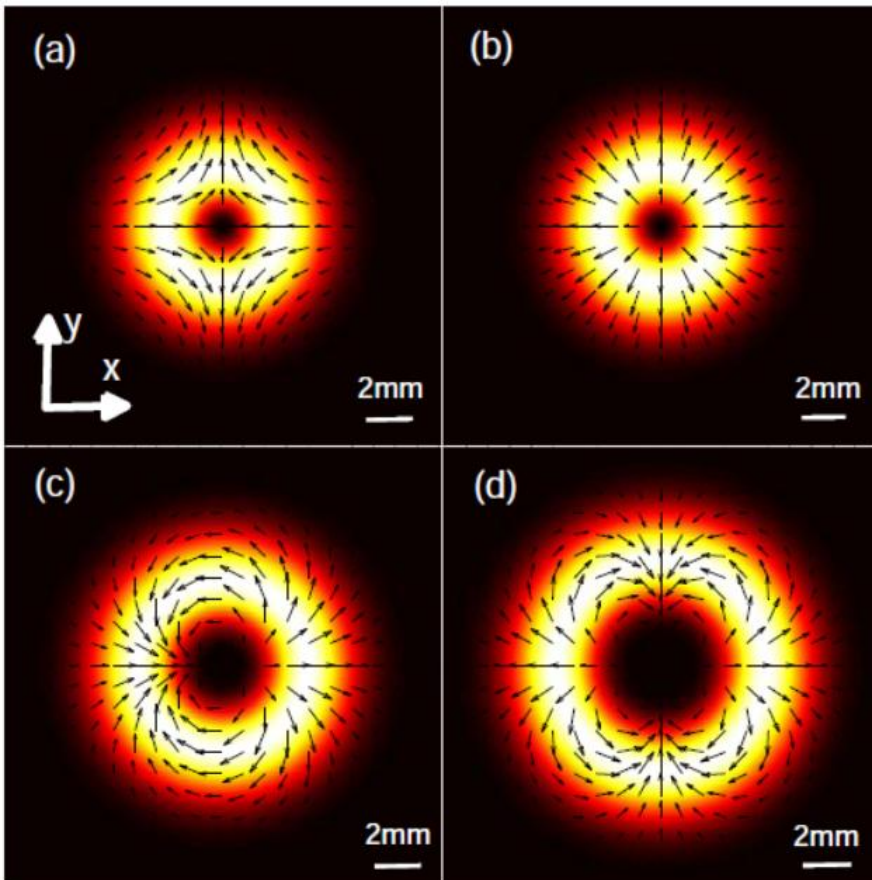
$$e^{ik[z \cos\theta + r \sin\theta \cos(\varphi - \phi')]} \sin\theta d\varphi$$

$$= \frac{-il_0 f}{\lambda} \int_0^{\theta_{\max}} d\theta \int_0^{2\pi} P(\theta) \cdot \mathbf{\Gamma} \cdot e^{ik[z \cos\theta + r \sin\theta \cos(\varphi - \phi')]} \sin\theta d\varphi$$

$$\mathbf{\Gamma} = \begin{pmatrix} [\xi(fg(\theta), \varphi) \cdot \cos\theta \cos\varphi - \eta(fg(\theta), \varphi) \cdot \sin\varphi] \bar{\mathbf{e}}_x \\ [\xi(fg(\theta), \varphi) \cdot \cos\theta \sin\varphi + \eta(fg(\theta), \varphi) \cdot \cos\varphi] \bar{\mathbf{e}}_y \\ [\xi(fg(\theta), \varphi) \cdot \sin\theta] \bar{\mathbf{e}}_z \end{pmatrix}$$

6.1. 物镜的强聚焦及衍射极限

2) 柱矢量光束的聚焦



$$\xi(\rho, \phi) = \cos[(n-1)\phi + \phi_0]$$

$$\eta(\rho, \phi) = \sin[(n-1)\phi + \phi_0]$$

$$\Gamma = \begin{pmatrix} \{\cos[(n-1)\phi + \phi_0] \cdot \cos \theta \cos \phi - \sin[(n-1)\phi + \phi_0] \cdot \sin \phi\} \bar{\mathbf{e}}_x \\ \{\cos[(n-1)\phi + \phi_0] \cdot \cos \theta \sin \phi + \sin[(n-1)\phi + \phi_0] \cdot \cos \phi\} \bar{\mathbf{e}}_y \\ \{\cos[(n-1)\phi + \phi_0] \cdot \sin \theta\} \bar{\mathbf{e}}_z \end{pmatrix}$$

极坐标形式:

$$\Gamma = \begin{pmatrix} \{\cos \theta \cos[(n-1)\phi + \phi_0] \cdot \cos(\phi - \phi') - \sin[(n-1)\phi + \phi_0] \cdot \sin(\phi - \phi')\} \bar{\mathbf{e}}_r \\ \{\cos \theta \cos[(n-1)\phi + \phi_0] \cdot \sin(\phi - \phi') + \sin[(n-1)\phi + \phi_0] \cdot \cos(\phi - \phi')\} \bar{\mathbf{e}}_{\phi'} \\ \{\cos[(n-1)\phi + \phi_0] \cdot \sin \theta\} \bar{\mathbf{e}}_z \end{pmatrix}$$

$$\cos(n\phi) \mathbf{e}_x + \sin(n\phi) \mathbf{e}_y$$

$$\mathbf{F}(r, \phi, z) = A(r, z) \{ \cos[(n-1)\phi + \phi_0] \hat{\mathbf{e}}_r + \sin[(n-1)\phi + \phi_0] \hat{\mathbf{e}}_{\phi} \}$$

6.1. 物镜的强聚焦及衍射极限

2) 柱矢量光束的聚焦

$$\mathbf{E}(r, \phi', z) = \frac{-il_0 f}{\lambda} \int_0^{\theta_{\max}} d\theta \int_0^{2\pi} P(\theta) \cdot \Gamma \cdot e^{ik[z \cos \theta + r \sin \theta \cos(\phi - \phi')]} \sin \theta d\phi$$

$$\Gamma = \begin{pmatrix} \{\cos \theta \cos[(n-1)\phi + \phi_0] \cdot \cos(\phi - \phi') - \sin[(n-1)\phi + \phi_0] \cdot \sin(\phi - \phi')\} \bar{\mathbf{e}}_r \\ \{\cos \theta \cos[(n-1)\phi + \phi_0] \cdot \sin(\phi - \phi') + \sin[(n-1)\phi + \phi_0] \cdot \cos(\phi - \phi')\} \bar{\mathbf{e}}_{\phi'} \\ \{\cos[(n-1)\phi + \phi_0] \cdot \sin \theta\} \bar{\mathbf{e}}_z \end{pmatrix}$$

$$\cos[(n-1)\phi + \phi_0] \cdot \cos(\phi - \phi') = \frac{1}{2} \cos[n\phi + \phi_0 - \phi'] + \frac{1}{2} \cos[(n-2)\phi + \phi_0 + \phi'],$$

$$\sin[(n-1)\phi + \phi_0] \cdot \sin(\phi - \phi') = -\frac{1}{2} \cos[n\phi + \phi_0 - \phi'] + \frac{1}{2} \cos[(n-2)\phi + \phi_0 + \phi']$$

$$\int_0^{2\pi} \cos(m\phi) e^{ikr \sin \theta \cos \phi} d\phi = 2\pi i^m J_m(kr \sin \theta),$$

$$\int_0^{2\pi} \sin(m\phi) e^{ikr \sin \theta \cos \phi} d\phi = 0,$$

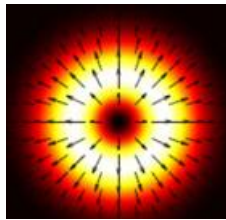
$$\mathbf{E}(r, \phi', z) = -i^{n+1} A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} \cos[(n-1)\phi' + \phi_0] \cdot [\cos \theta (J_n - J_{n-2}) + (J_n + J_{n-2})] \bar{\mathbf{e}}_r \\ \sin[(n-1)\phi' + \phi_0] \cdot [\cos \theta (J_n + J_{n-2}) + (J_n - J_{n-2})] \bar{\mathbf{e}}_{\phi'} \\ - 2i \cos[(n-1)\phi' + \phi_0] \sin \theta \cdot J_{n-1} \bar{\mathbf{e}}_z \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$$

6.1. 物镜的强聚焦及衍射极限

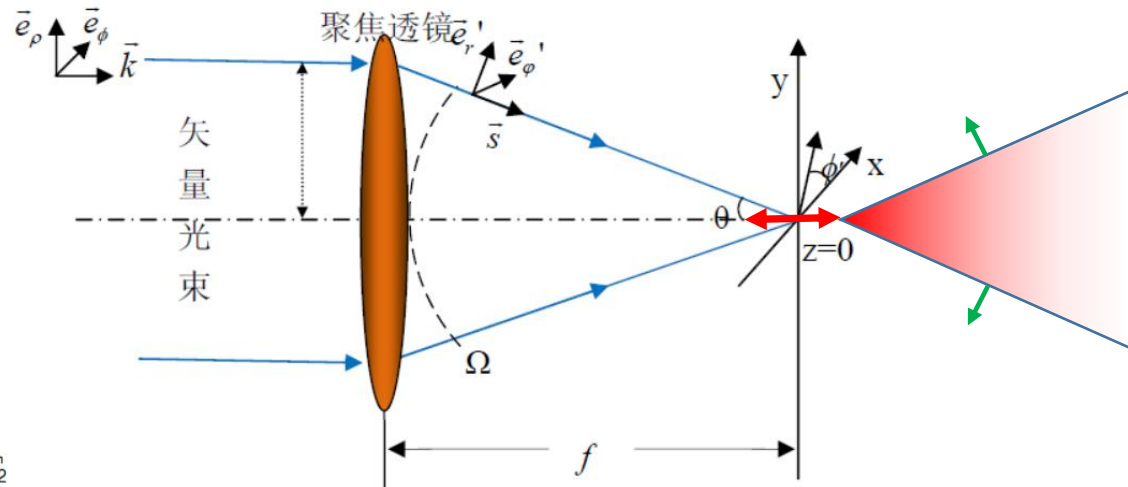
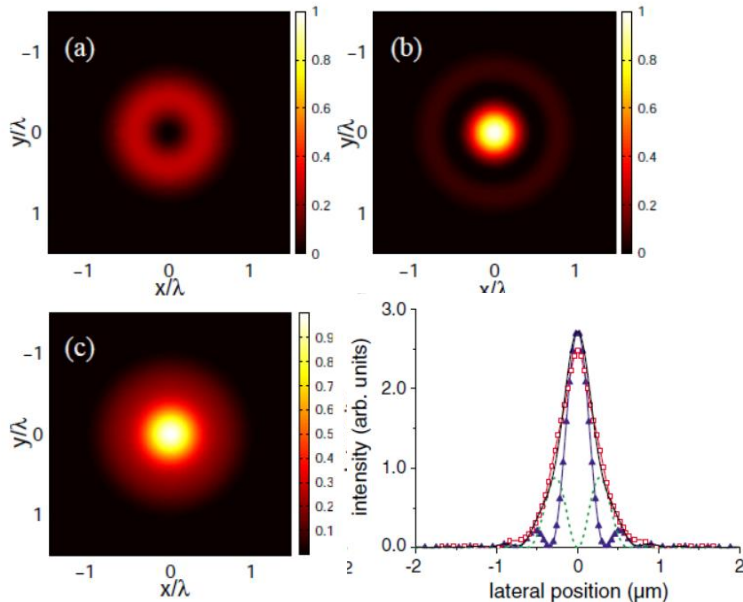
2) 柱矢量光束的聚焦

$$\mathbf{E}(r, \phi', z) = -i^{n+1} A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} \cos[(n-1)\phi' + \varphi_0] \cdot [\cos \theta (J_n - J_{n-2}) + (J_n + J_{n-2})] \bar{\mathbf{e}}_r \\ \sin[(n-1)\phi' + \varphi_0] \cdot [\cos \theta (J_n + J_{n-2}) + (J_n - J_{n-2})] \bar{\mathbf{e}}_{\phi'} \\ - 2i \cos[(n-1)\phi' + \varphi_0] \sin \theta \cdot J_{n-1} \bar{\mathbf{e}}_z \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$$

若 $n=1$ 和 $\varphi_0=0$



$$\text{则 } \mathbf{E}_{\text{rad}}(r, \phi', z) = A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} 2 \cos \theta \cdot J_1 \bar{\mathbf{e}}_r \\ 0 \\ - 2i \sin \theta \cdot J_0 \bar{\mathbf{e}}_z \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$$

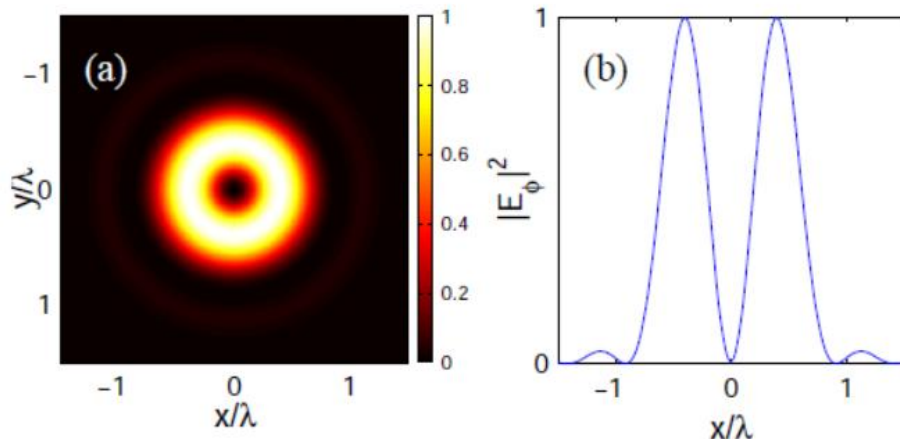
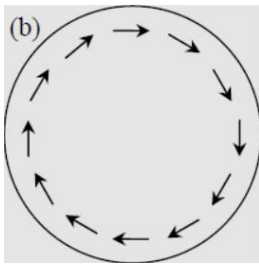


6.1. 物镜的强聚焦及衍射极限

2) 柱矢量光束的聚焦

$$\mathbf{E}(r, \phi', z) = -i^{n+1} A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} \cos[(n-1)\phi' + \phi_0] \cdot [\cos \theta (J_n - J_{n-2}) + (J_n + J_{n-2})] \bar{\mathbf{e}}_r \\ \sin[(n-1)\phi' + \phi_0] \cdot [\cos \theta (J_n + J_{n-2}) + (J_n - J_{n-2})] \bar{\mathbf{e}}_{\phi'} \\ - 2i \cos[(n-1)\phi' + \phi_0] \sin \theta \cdot J_{n-1} \bar{\mathbf{e}}_z \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$$

若 $n=1$ 和 $\phi_0 = \frac{\pi}{2}$ 则 $\mathbf{E}_{azi}(r, \phi', z) = A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} 0 \\ 2J_1 \bar{\mathbf{e}}_{\phi'} \\ 0 \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$

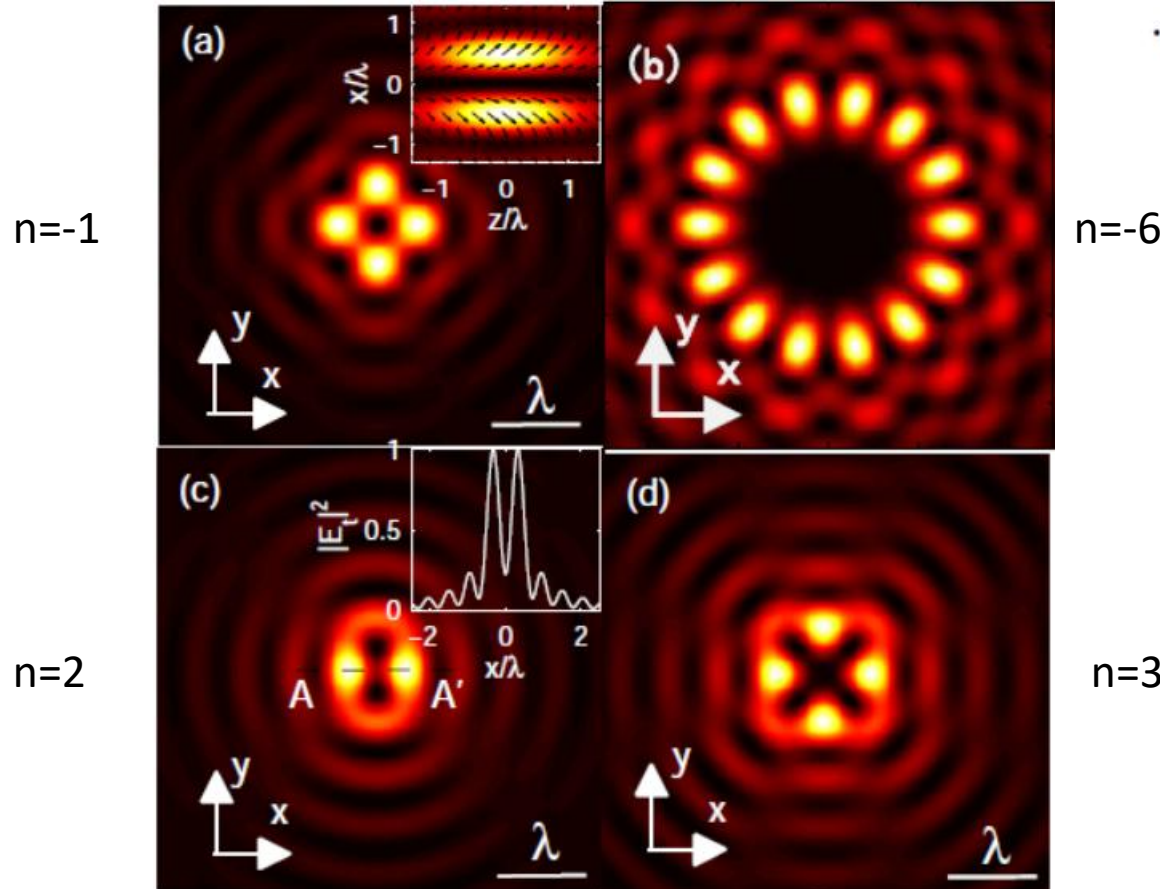


如何利用角向偏振光实现“小焦斑”？

6.1. 物镜的强聚焦及衍射极限

2) 柱矢量光束的聚焦

$$\mathbf{E}(r, \phi', z) = -i^{n+1} A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} \cos[(n-1)\phi' + \phi_0] \cdot [\cos \theta (J_n - J_{n-2}) + (J_n + J_{n-2})] \bar{\mathbf{e}}_r \\ \sin[(n-1)\phi' + \phi_0] \cdot [\cos \theta (J_n + J_{n-2}) + (J_n - J_{n-2})] \bar{\mathbf{e}}_{\phi'} \\ - 2i \cos[(n-1)\phi' + \phi_0] \sin \theta \cdot J_{n-1} \bar{\mathbf{e}}_z \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$$



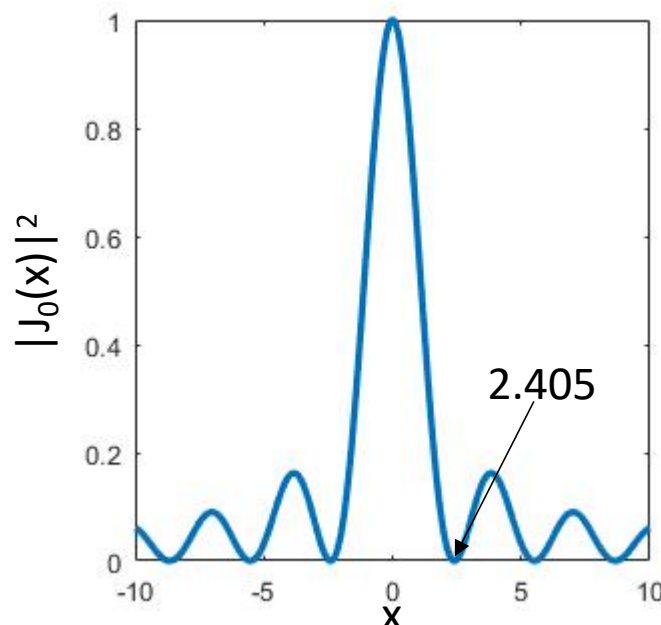
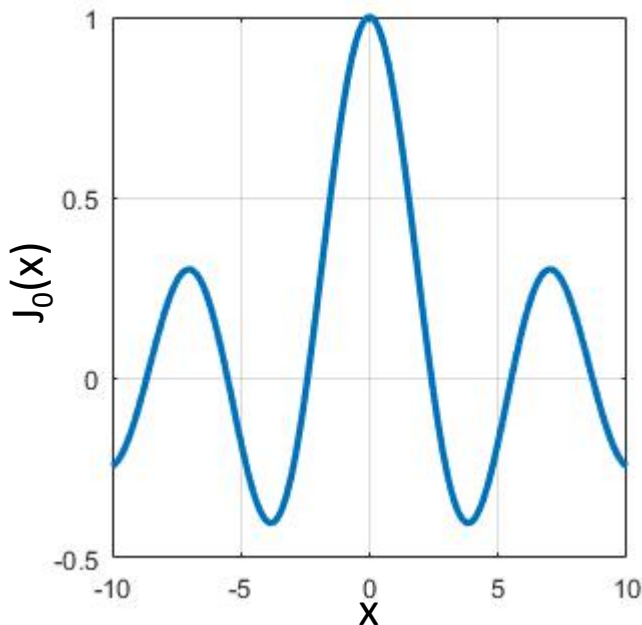
6.1. 物镜的强聚焦及衍射极限

3) 物镜的衍射极限

径向偏振光的聚焦

$$\mathbf{E}_{rad}(r, \phi', z) = A \int_0^{\theta_{\max}} P(\theta) \cdot \begin{pmatrix} 2 \cos \theta \cdot J_1 \bar{\mathbf{e}}_r \\ 0 \\ -2i \sin \theta \cdot J_0 \bar{\mathbf{e}}_z \end{pmatrix} \cdot e^{ikz \cos \theta} \sin \theta d\theta$$

$$\theta \rightarrow \frac{\pi}{2} \quad \cos \theta = 0, \sin \theta = 1 \rightarrow I = |\mathbf{E}|^2 = |J_0(krNA)|^2$$



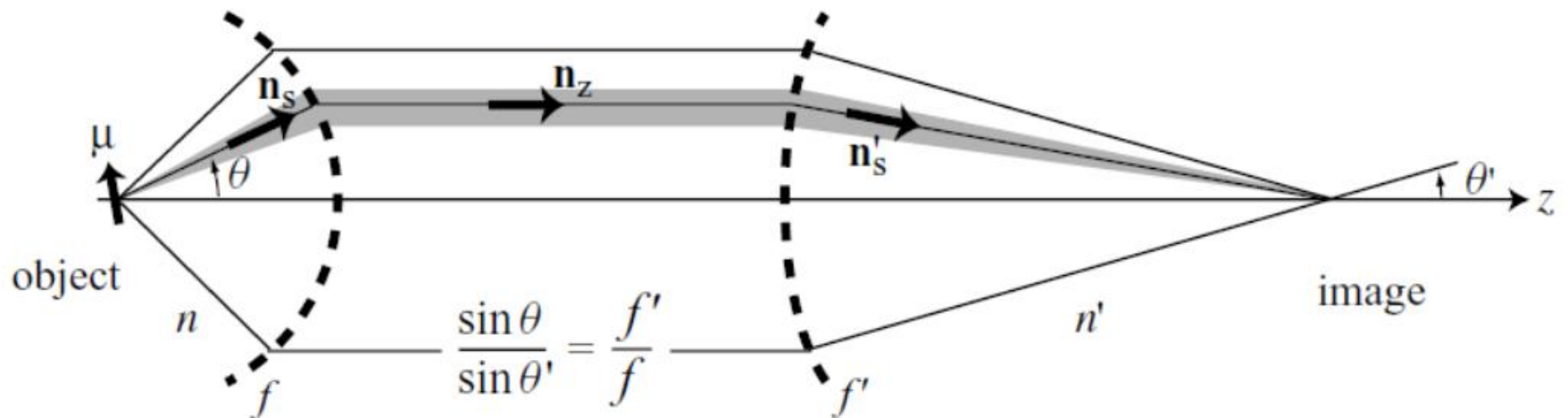
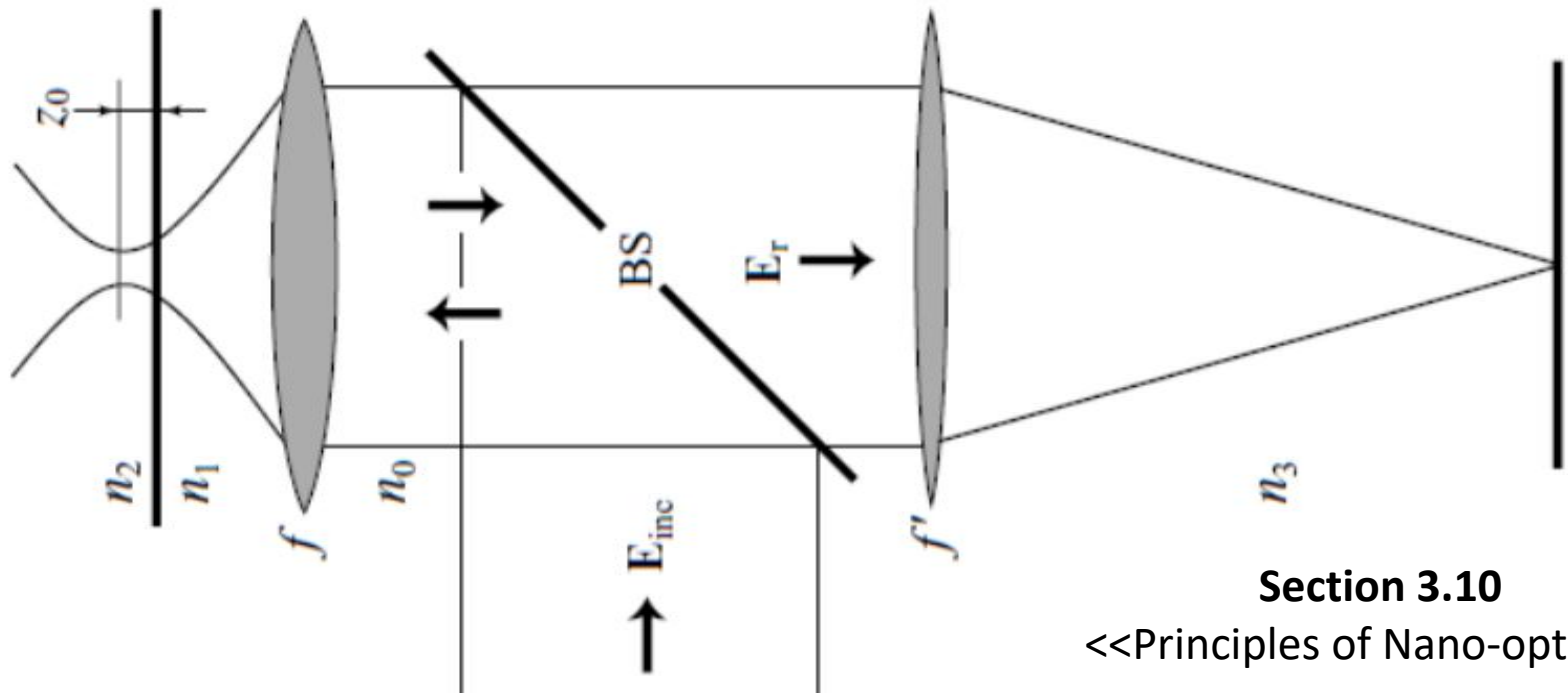
$$krNA = 2.405$$



$$r = 0.38\lambda/NA$$

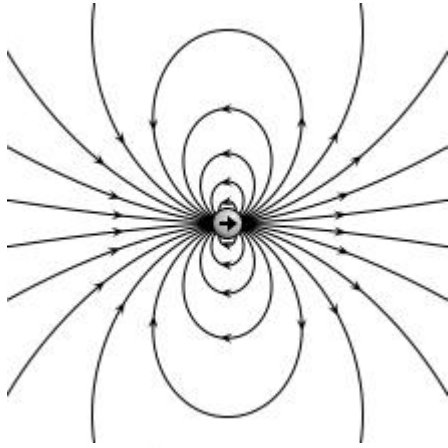
物镜聚焦的真实衍射极限!

6.2. 物镜成像的矢量波动理论



6.2. 物镜成像的矢量波动理论

电偶极子的远场辐射



$$\mathbf{p} = \int \mathbf{x}' \rho(\mathbf{x}') d^3x'$$

电偶极子辐射电场和磁场:

$$\mathbf{H} = \frac{ck^2}{4\pi} (\mathbf{n} \times \mathbf{p}) \frac{e^{ikr}}{r} \left(1 - \frac{1}{ikr}\right)$$

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \left\{ \underbrace{k^2(\mathbf{n} \times \mathbf{p}) \times \mathbf{n} \frac{e^{ikr}}{r}}_{\text{远场项}} + \underbrace{[3\mathbf{n}(\mathbf{n} \cdot \mathbf{p}) - \mathbf{p}]\left(\frac{1}{r^3} - \frac{ik}{r^2}\right)e^{ikr}}_{\text{近场项}} \right\}$$

远场项

近场项

电偶极子的远场辐射电场:

$$\mathbf{E} = -\omega^2 \mu_0 [\mathbf{r} \times \mathbf{r} \times \boldsymbol{\mu}] \exp(ikr) / 4\pi r^3$$

$$= -\omega^2 \mu_0 \exp(ikr) / r \cdot \begin{bmatrix} (1 - x^2/r^2) & -xy/r^2 & -xz/r^2 \\ -xy/r^2 & (1 - y^2/r^2) & -yz/r^2 \\ -xz/r^2 & -yz/r^2 & (1 - z^2/r^2) \end{bmatrix} \begin{bmatrix} \mu_x \\ \mu_y \\ \mu_z \end{bmatrix}$$

利用球坐标,

$$z = r \cdot \cos\theta$$

$$x = r \cdot \sin\theta \cdot \cos\varphi$$

$$y = r \cdot \sin\theta \cdot \sin\varphi$$

$$\mathbf{H} = \frac{1}{\mu_0} \nabla \times \mathbf{A}$$

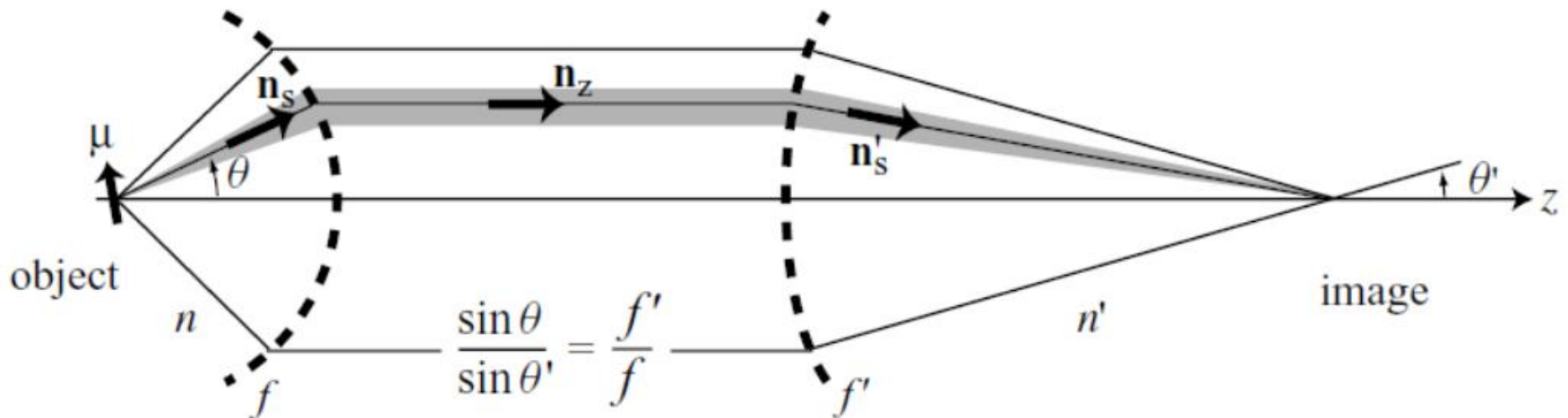
$$\mathbf{E} = \frac{iZ_0}{k} \nabla \times \mathbf{H}$$

6.2. 物镜成像的矢量波动理论

$$\mathbf{E}(\mathbf{r}) = \frac{\omega^2}{\varepsilon_0 c^2} \vec{\mathbf{G}}(\mathbf{r}, \mathbf{r}_0) \cdot \boldsymbol{\mu}$$

$$\vec{\mathbf{G}}_{\infty}(\mathbf{r}, 0) = \frac{\exp(ikr)}{4\pi r}$$

$$\times \begin{bmatrix} (1 - \cos^2\phi \sin^2\theta) & -\sin\phi \cos\phi \sin^2\theta & -\cos\phi \sin\theta \cos\theta \\ -\sin\phi \cos\phi \sin^2\theta & (1 - \sin^2\phi \sin^2\theta) & -\sin\phi \sin\theta \cos\theta \\ -\cos\phi \sin\theta \cos\theta & -\sin\phi \sin\theta \cos\theta & \sin^2\theta \end{bmatrix}$$



6.2. 物镜成像的矢量波动理论

如果电偶极子是沿x方向，则

$$\mathbf{E}(\mathbf{r}) = \frac{\omega^2}{\varepsilon_0 c^2} \hat{\mathbf{G}}(\mathbf{r}, 0) \cdot \begin{bmatrix} \mu_x \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{\omega^2}{\varepsilon_0 c^2} \frac{\exp(ikr)}{4\pi r} \mu_x \begin{bmatrix} (1 - \cos^2\phi \sin^2\theta) \\ -\sin\phi \cos\phi \sin^2\theta \\ -\cos\phi \sin\theta \cos\theta \end{bmatrix}$$

$$\mathbf{n}_\theta = \cos\theta(\cos\phi\mathbf{e}_x + \sin\phi\mathbf{e}_y) - \sin\theta\mathbf{e}_z$$

$$\mathbf{n}_r = \cos\phi\mathbf{e}_x + \sin\phi\mathbf{e}_y$$

$$\mathbf{n}_{\theta'} = \cos\theta'(\cos\phi\mathbf{e}_x + \sin\phi\mathbf{e}_y) + \sin\theta'\mathbf{e}_z$$

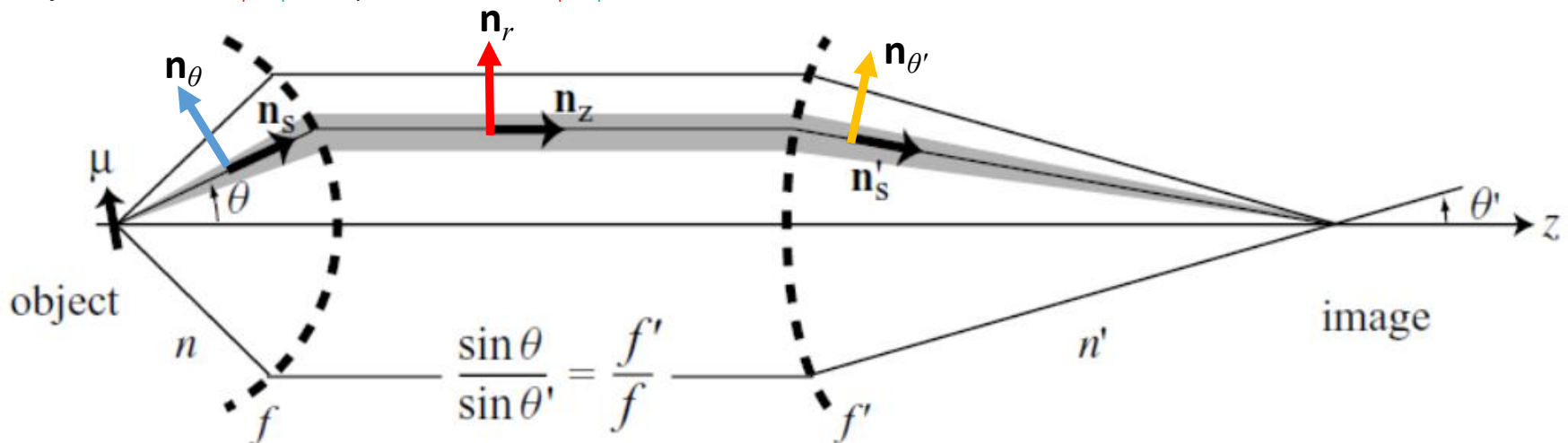
$$\mathbf{n}_\phi = -\sin\phi\mathbf{e}_x + \cos\phi\mathbf{e}_y$$

$$P(\theta) = P(\rho) \sqrt{g(\theta)g'(\theta)/\sin\theta}$$

$$\mathbf{E}_r = \cos\phi\cos\theta \quad \mathbf{E}_\phi = -\sin\phi$$

$$\mathbf{E}_{\text{obj}}(\theta) = \mathbf{E}_r \mathbf{n}_\theta + \mathbf{E}_\phi \mathbf{n}_\phi \quad \mathbf{E}_p(\mathbf{r}) = (\mathbf{E}_r \mathbf{n}_r + \mathbf{E}_\phi \mathbf{n}_\phi) / (\cos\theta)^{1/2}$$

$$\mathbf{E}_{\text{image}}(\theta') = (\mathbf{E}_r \mathbf{n}_{\theta'} + \mathbf{E}_\phi \mathbf{n}_\phi) \cdot (\cos\theta'/\cos\theta)^{1/2}$$



6.2. 物镜成像的矢量波动理论

$$\mathbf{E}_{\text{image}}(\theta') = (\mathbf{E}_r \mathbf{n}_{\theta'} + \mathbf{E}_\phi \mathbf{n}_\phi) \cdot (\cos\theta'/\cos\theta)^{1/2}$$

$$\mathbf{n}_{\theta'} = \cos\theta'(\cos\phi\mathbf{e}_x + \sin\phi\mathbf{e}_y) + \sin\theta'\mathbf{e}_z$$

$$\mathbf{n}_\phi = -\sin\phi\mathbf{e}_x + \cos\phi\mathbf{e}_y$$

For μ_x ,

$$\mathbf{E}_r = \cos\phi\cos\theta \quad \mathbf{E}_\phi = -\sin\phi$$

$$\mathbf{E}_{\text{image}}(\theta') = \begin{bmatrix} (1 + \cos\theta\cos\theta') - (1 - \cos\theta\cos\theta')\cos 2\phi \\ -(1 - \cos\theta\cos\theta')\sin 2\phi \\ 2\cos\theta\sin\theta'\cos\phi \end{bmatrix} \cdot (\cos\theta'/\cos\theta)^{1/2}$$

For μ_y ,

$$\mathbf{E}_r = \sin\phi\cos\theta \quad \mathbf{E}_\phi = \cos\phi$$

$$\mathbf{E}_{\text{image}}(\theta') = \begin{bmatrix} -(1 - \cos\theta\cos\theta')\sin 2\phi \\ (1 + \cos\theta\cos\theta') + (1 - \cos\theta\cos\theta')\cos 2\phi \\ 2\cos\theta\sin\theta'\sin\phi \end{bmatrix} \cdot (\cos\theta'/\cos\theta)^{1/2}$$

For μ_z ,

$$\mathbf{E}_r = -\sin\theta \quad \mathbf{E}_\phi = 0$$

$$\mathbf{E}_{\text{image}}(\theta') = \begin{bmatrix} -2\sin\theta\cos\theta'\cos\phi \\ -2\sin\theta\cos\theta'\sin\phi \\ -2\sin\theta\sin\theta' \end{bmatrix} \cdot (\cos\theta'/\cos\theta)^{1/2}$$

$$\mathbf{E}_{\text{image}}(\theta') = \frac{\omega^2}{\varepsilon_0 c^2} \frac{\exp(ikf)}{8\pi f} \mathbf{G}' \cdot \boldsymbol{\mu}$$

$$\boldsymbol{\mu} = \begin{bmatrix} \mu_x \\ \mu_y \\ \mu_z \end{bmatrix}$$

$$\mathbf{G}' = \sqrt{\frac{\cos\theta'}{\cos\theta}} \times$$

$$\begin{bmatrix} (1 + \cos\theta\cos\theta') - (1 - \cos\theta\cos\theta')\cos 2\phi & -(1 - \cos\theta\cos\theta')\sin 2\phi & -2\sin\theta\cos\theta'\cos\phi \\ -(1 - \cos\theta\cos\theta')\sin 2\phi & (1 + \cos\theta\cos\theta') + (1 - \cos\theta\cos\theta')\cos 2\phi & -2\sin\theta\cos\theta'\sin\phi \\ 2\cos\theta\sin\theta'\cos\phi & 2\cos\theta\sin\theta'\sin\phi & -2\sin\theta\sin\theta' \end{bmatrix}$$

6.2. 物镜成像的矢量波动理论

$$\mathbf{E}_{\text{image}}(\theta') = \frac{\omega^2}{\varepsilon_0 c^2} \frac{\exp(ikf)}{8\pi f} \mathbf{G}' \cdot \boldsymbol{\mu}$$

$$\mathbf{E}(\rho, \varphi, z) = \frac{ik f' e^{-ik f'}}{2\pi} \int_0^{\theta_{\max}} \int_0^{2\pi} \mathbf{E}_{\text{image}}(\theta') e^{-ik z \cos \theta'} e^{ik \rho \sin \theta' \cos(\phi - \varphi)} \sin \theta' d\phi d\theta'$$

$$\frac{\sin \theta}{\sin \theta'} = \frac{f'}{f} \quad \cos \theta' = \sqrt{1 - (f/f')^2 \sin^2 \theta} \quad [1 \pm (f/f')^2 \sin^2 \theta]^{1/n} \approx 1 \pm \frac{1}{n} \left(\frac{f}{f'} \right)^2 \sin^2 \theta$$

$$\mathbf{E}(\rho, \varphi, z) = \frac{ik f' e^{-ik f'}}{2\pi} \frac{f^2}{f'^2} \int_0^{\theta_{\max}} \int_0^{2\pi} \mathbf{E}_{\text{image}}(\theta') e^{(i/2)k z (f/f')^2 \sin^2 \theta} \times e^{ik \rho (f/f') \sin \theta \cos(\phi - \varphi)} \sin \theta \cos \theta d\phi d\theta,$$

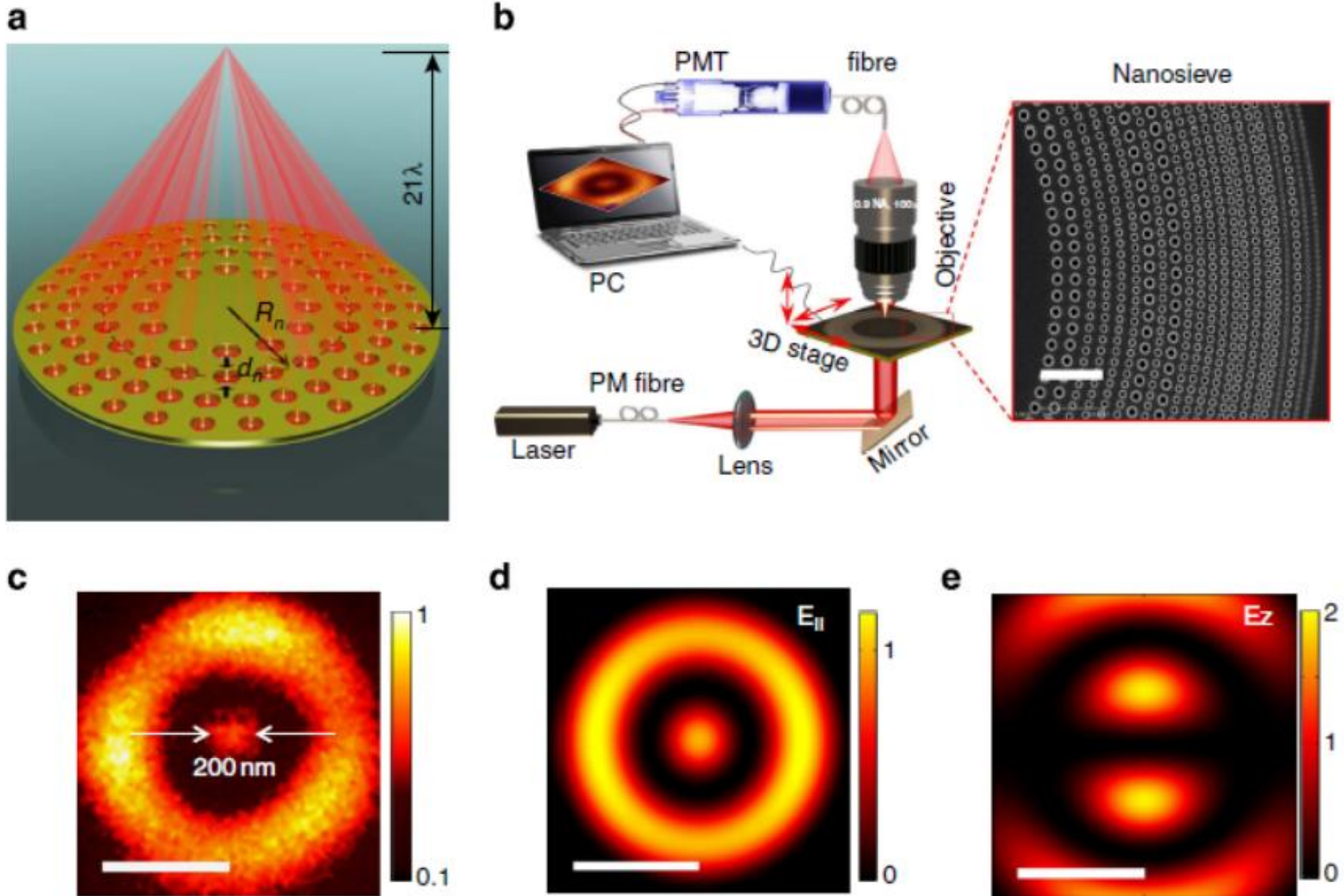
$$\mathbf{G}' = \sqrt{\frac{\cos \theta'}{\cos \theta}} \times$$

$$\cos \theta' = 1, \quad \sin \theta' = \sin \theta / (f'/f) = \sin \theta / M \rightarrow 0$$

结论1: E_z 为零!!!

$$\begin{bmatrix} (1 + \cos \theta \cos \theta') - (1 - \cos \theta \cos \theta') \cos 2\phi & -(1 - \cos \theta \cos \theta') \sin 2\phi & -2 \sin \theta \cos \theta' \cos \phi \\ -(1 - \cos \theta \cos \theta') \sin 2\phi & (1 + \cos \theta \cos \theta') + (1 - \cos \theta \cos \theta') \cos 2\phi & -2 \sin \theta \cos \theta' \sin \phi \\ 2 \cos \theta \sin \theta' \cos \phi & 2 \cos \theta \sin \theta' \sin \phi & -2 \sin \theta \sin \theta' \end{bmatrix}$$

6.2. 物镜成像的矢量波动理论



6.2. 物镜成像的矢量波动理论

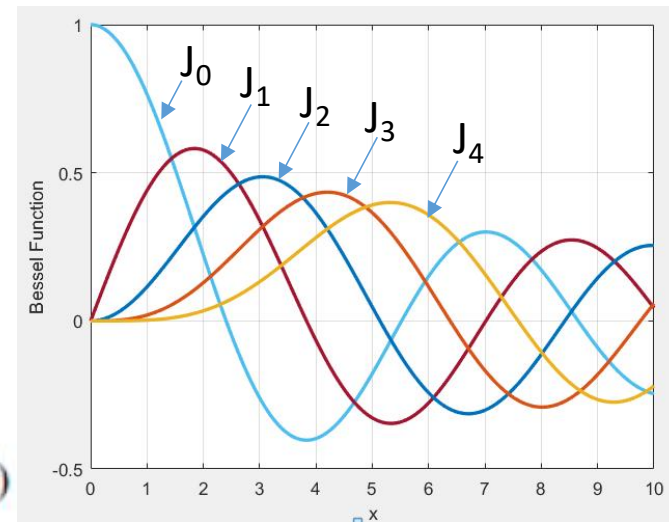
$$\mathbf{E}(\rho, \varphi, z) = \frac{\omega^2}{\varepsilon_0 c^2} \vec{\mathbf{G}}_{\text{PSF}}(\rho, \varphi, z) \cdot \boldsymbol{\mu}$$

$$\vec{\mathbf{G}}_{\text{PSF}} = \frac{k'}{8\pi i} \frac{f}{f'} e^{i(kf - k'f')} \begin{bmatrix} (\tilde{I}_{00} + \tilde{I}_{02} \cos 2\varphi) & \tilde{I}_{02} \sin 2\varphi & -2i\tilde{I}_{01} \cos \varphi \\ \tilde{I}_{02} \sin 2\varphi & (\tilde{I}_{00} - \tilde{I}_{02} \cos 2\varphi) & -2i\tilde{I}_{01} \sin \varphi \\ 0 & 0 & 0 \end{bmatrix} \sqrt{\frac{n}{n'}}$$

$$\begin{aligned} \tilde{I}_{00}(\rho, z) &= \int_0^{\theta_{\max}} (\cos \theta)^{1/2} \sin \theta (1 + \cos \theta) J_0(k' \rho \sin \theta f/f') \\ &\quad \times \exp \{ik'z [1 - 1/2(f/f')^2 \sin^2 \theta]\} d\theta, \end{aligned}$$

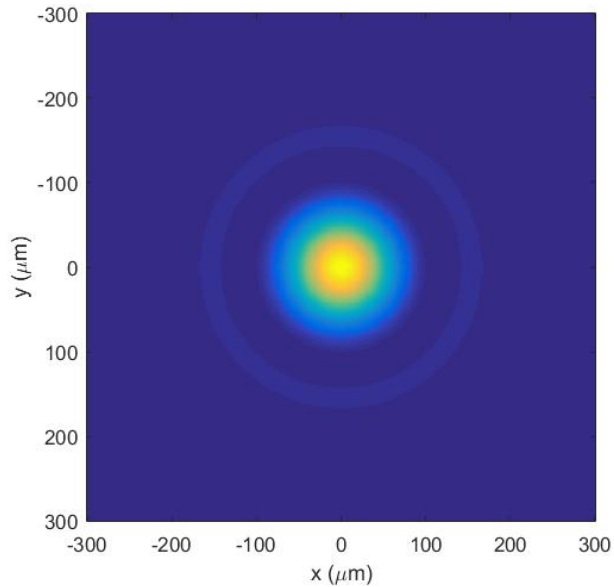
$$\begin{aligned} \tilde{I}_{01}(\rho, z) &= \int_0^{\theta_{\max}} (\cos \theta)^{1/2} \sin^2 \theta J_1(k' \rho \sin \theta f/f') \\ &\quad \times \exp \{ik'z [1 - 1/2(f/f')^2 \sin^2 \theta]\} d\theta, \end{aligned}$$

$$\begin{aligned} \tilde{I}_{02}(\rho, z) &= \int_0^{\theta_{\max}} (\cos \theta)^{1/2} \sin \theta (1 - \cos \theta) J_2(k' \rho \sin \theta f/f') \\ &\quad \times \exp \{ik'z [1 - 1/2(f/f')^2 \sin^2 \theta]\} d\theta. \end{aligned}$$

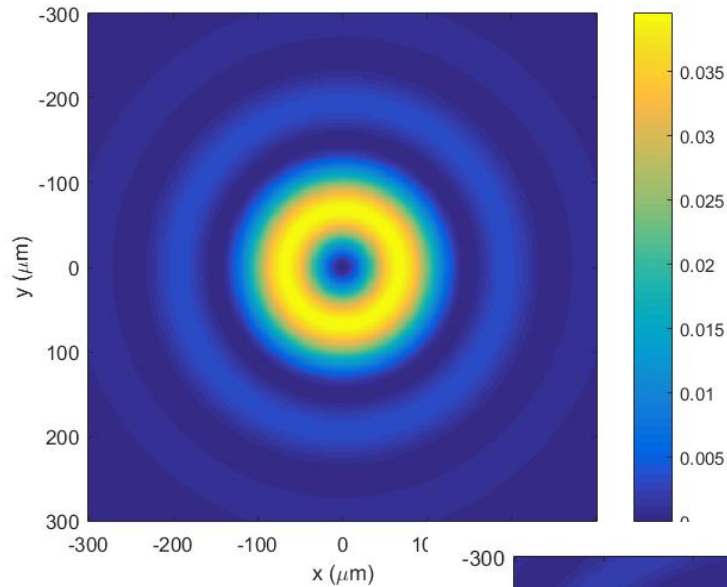


6.2. 物镜成像的矢量波动理论

$$\tilde{I}_{00}(\rho, z)$$

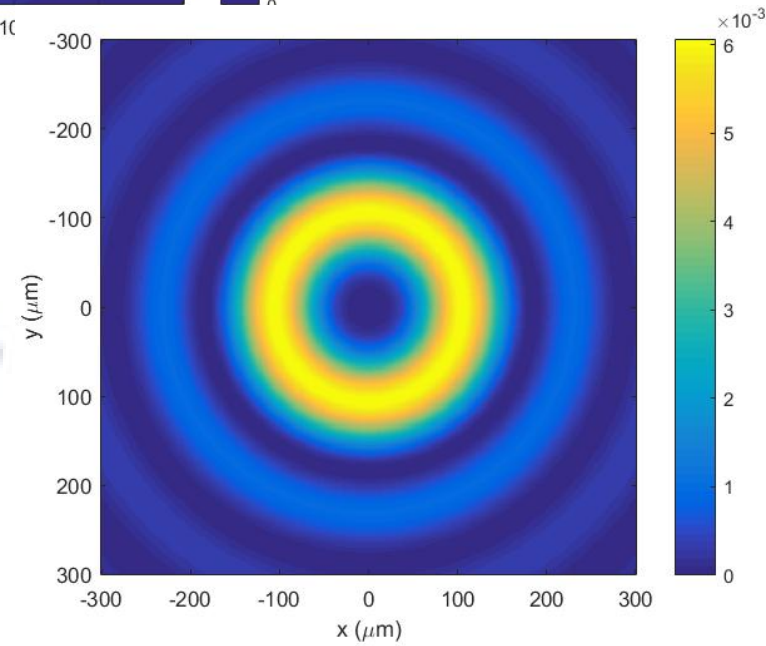


$$\tilde{I}_{01}(\rho, z)$$



$f=350\mu\text{m}$
 $f'=10\text{cm}$
 $\text{NA}=0.95$
Wavelength= 0.633
 $z=0$

$$\tilde{I}_{02}(\rho, z)$$

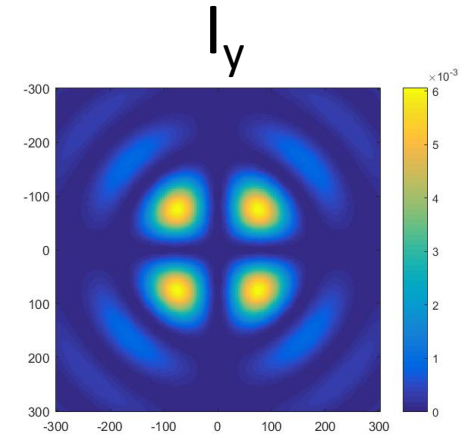
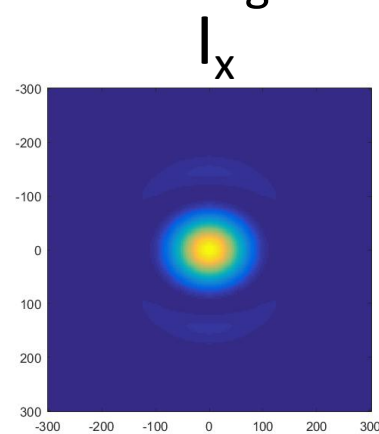
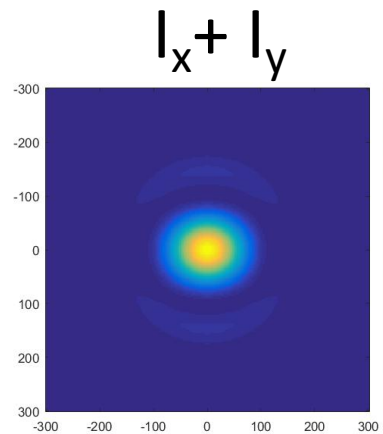


6.2. 物镜成像的矢量波动理论

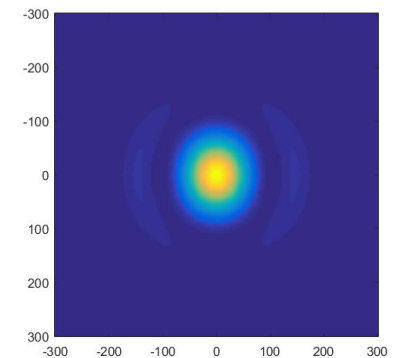
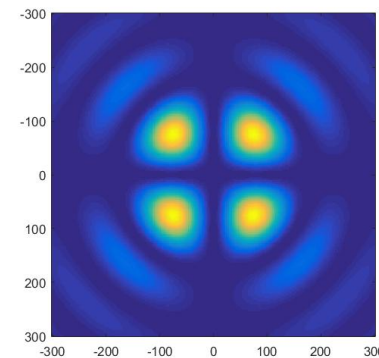
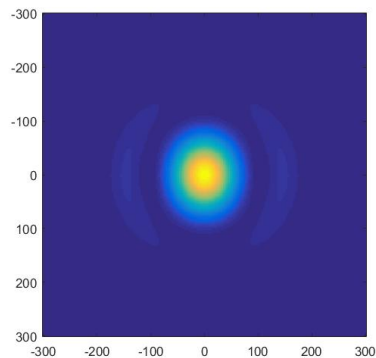
Objects

Images

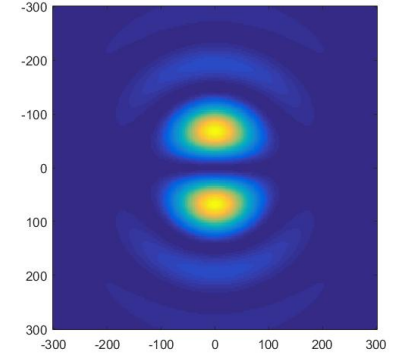
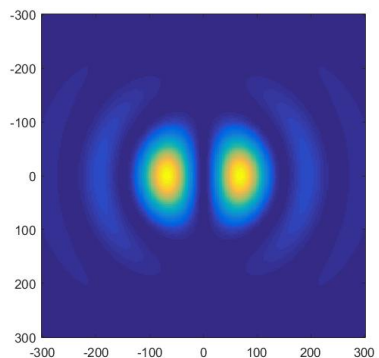
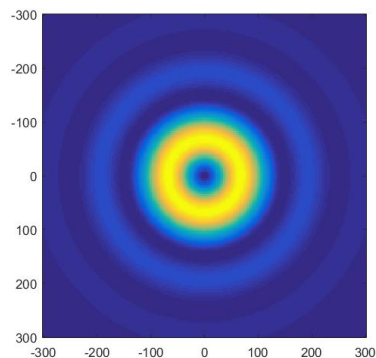
μ_x



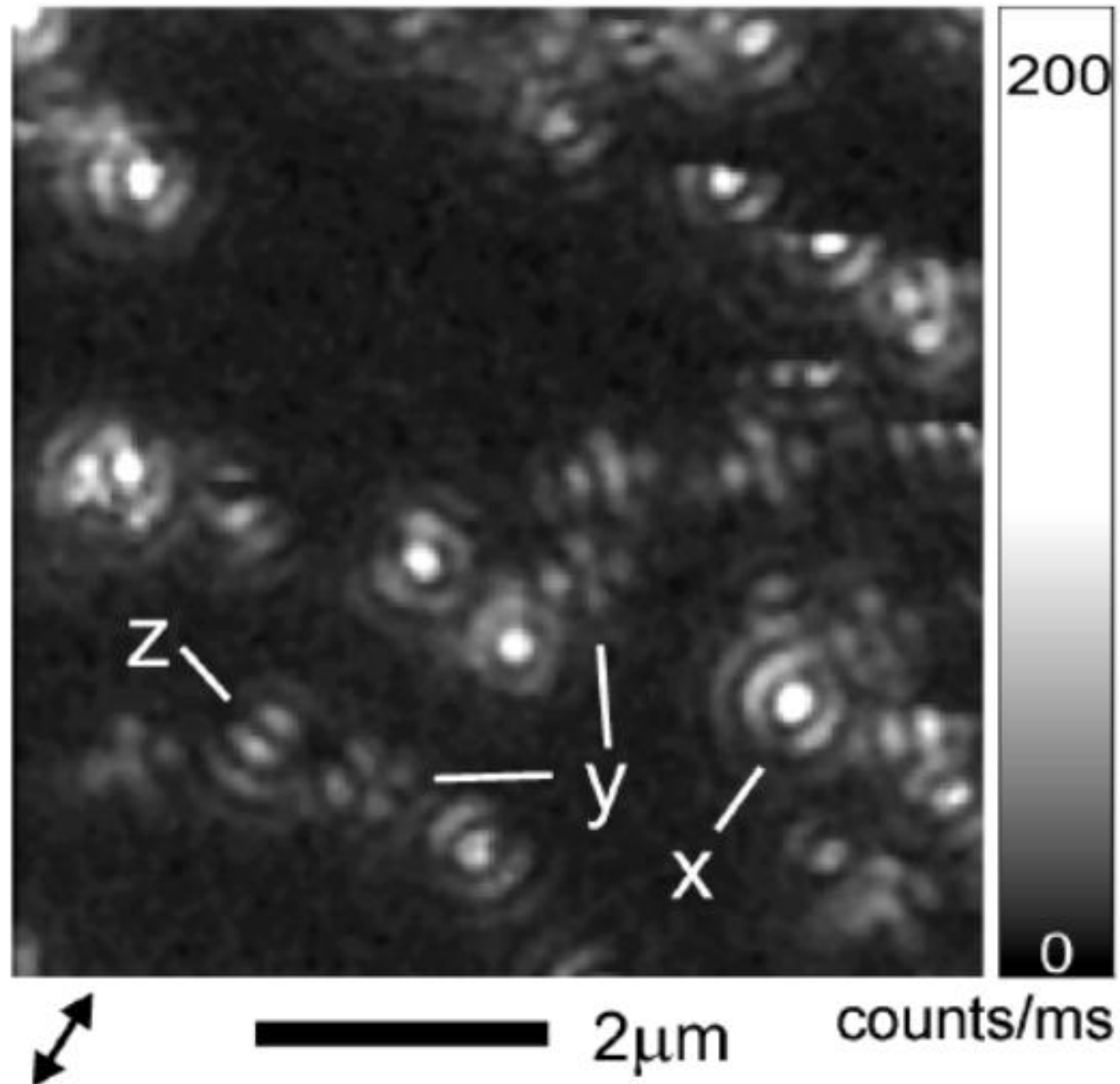
μ_y



μ_z



6.2. 物镜成像的矢量波动理论



6.2. 物镜成像的矢量波动理论

结论2: 在高数值孔径物镜的成像系统中, **x**和**y**方向电偶极子的点扩散函数近似为**Airy斑**, 这与低数值孔径透镜的聚焦类似。

$$\vec{G}_{\text{PSF}} = \frac{k'}{8\pi i} \frac{f}{f'} e^{i(kf - k'f')} \begin{bmatrix} (\tilde{I}_{00} + \tilde{I}_{02} \cos 2\varphi) & \tilde{I}_{02} \sin 2\varphi & -2i\tilde{I}_{01} \cos \varphi \\ \tilde{I}_{02} \sin 2\varphi & (\tilde{I}_{00} - \tilde{I}_{02} \cos 2\varphi) & -2i\tilde{I}_{01} \sin \varphi \\ 0 & 0 & 0 \end{bmatrix} \sqrt{\frac{n}{n'}}$$

对于低数值孔径透镜, $\theta_{\max}$ 很小

$$\begin{aligned} \tilde{I}_{00}(\rho, z) &= \int_0^{\theta_{\max}} (\cos \theta)^{1/2} \sin \theta (1 + \cos \theta) J_0(k' \rho \sin \theta f/f') \\ &\times \exp \{ik'z [1 - 1/2(f/f')^2 \sin^2 \theta]\} d\theta, \end{aligned}$$

$$\cos \theta \rightarrow 1 \quad \sin \theta \rightarrow \theta$$

$$I_{00} \approx \int_0^{\theta_{\max}} \theta J_0(k\rho\theta/M) d\theta$$

$$x = k\rho\theta/M$$

$$\begin{aligned} \tilde{I}_{01}(\rho, z) &= \int_0^{\theta_{\max}} (\cos \theta)^{1/2} \sin^2 \theta J_1(k' \rho \sin \theta f/f') \\ &\times \exp \{ik'z [1 - 1/2(f/f')^2 \sin^2 \theta]\} d\theta, \end{aligned}$$

$$= \left(\frac{M}{k\rho}\right)^2 \int_0^{k\rho\theta_{\max}/M} x J_0(x) dx$$

$$\begin{aligned} \tilde{I}_{02}(\rho, z) &= \int_0^{\theta_{\max}} (\cos \theta)^{1/2} \sin \theta (1 - \cos \theta) J_2(k' \rho \sin \theta f/f') \\ &\times \exp \{ik'z [1 - 1/2(f/f')^2 \sin^2 \theta]\} d\theta. \end{aligned}$$

$$= \left(\frac{M}{k\rho}\right)^2 x J_1(x) \Big|_0^{k\rho\theta_{\max}/M}$$

$$h \propto \frac{J_1(k\rho\theta_{\max}/M)}{k\rho\theta_{\max}/M} = \frac{J_1(k\rho NA/M)}{k\rho NA/M}$$

6.2. 物镜成像的矢量波动理论

物镜聚焦和成像小结：

1. 矢量光束被高数值孔径物镜聚焦后，焦平面的光场也是矢量的，且产生了 E_z 分量。（6.1）

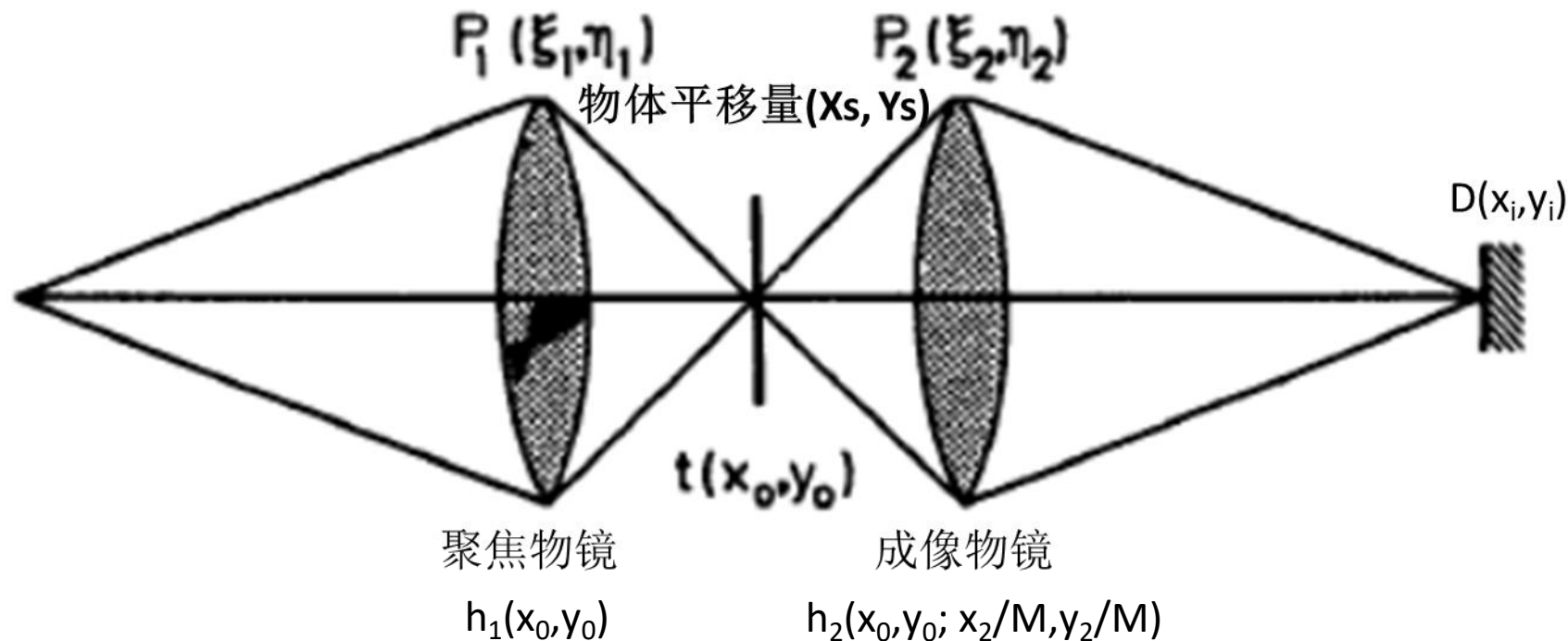
2. 被物镜会聚后的光场，或由原子、分子辐射的光，被高数值孔径物镜收集、准直，再被另外一个低数值孔径物镜成像后，**原来物空间的** E_z 分量，在像空间会严重衰减至零；只有电场的横向分量（ E_x 和 E_y ）才能被记录。

但是，物空间的 E_z 分量可以当作一个沿 z 轴辐射的电偶极子，其远场辐射光里包含 E_x 和 E_y 分量，所以，一个 z 向辐射的电偶极子，其image为径向偏振的环形光束。（6.2）

3. 物镜聚焦和成像都满足Rayleigh衍射极限： $0.61\lambda/NA$

如果只用高空间频率（与光轴间夹角大的光线）的光来聚焦或成像，其衍射极限为 $0.38\lambda/NA$ 。

6.3. 扫描共焦显微镜原理



$$U(x_i, y_i; x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(x_0, y_0) \cdot t_0(x_s - x_0, y_s - y_0) \cdot h_2(x_i, y_i; x_0, y_0) dx_0 dy_0$$

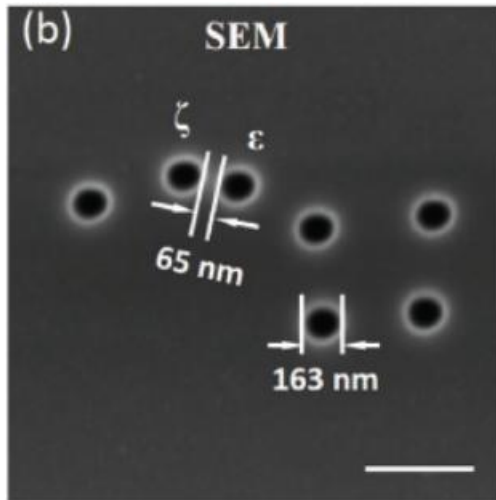
$$h_2(x_i, y_i; x_0, y_0) = \frac{D^2}{\lambda^2 d_0 d_i} \frac{J_1(t)}{t}$$

$$t = k \cdot NA \cdot \sqrt{(x_i/M - x_0)^2 + (y_i/M - y_0)^2}$$

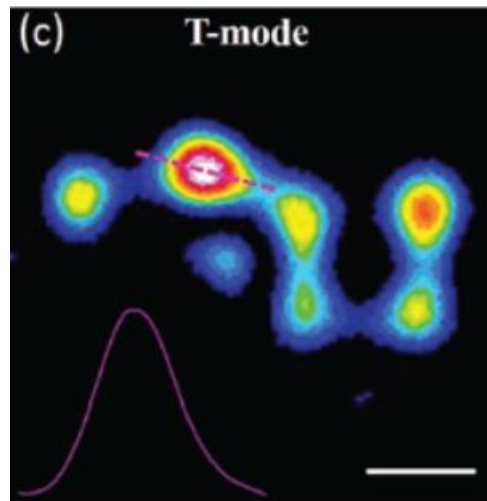
$$I(x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(x_i, y_i) \cdot |U(x_i, y_i; x_s, y_s)|^2 dx_i dy_i$$

6.3. 扫描共焦显微镜原理

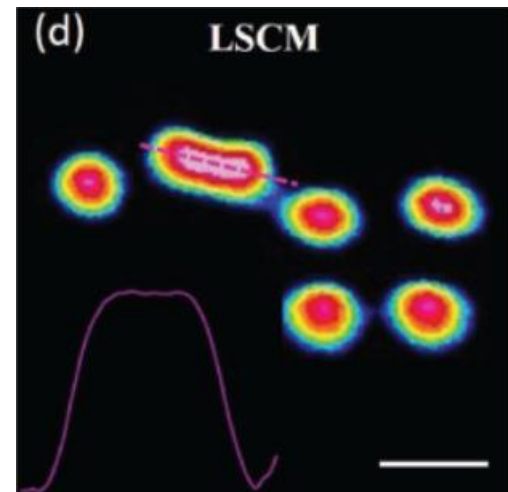
物体



物镜直接成像



扫描共焦成像



$$U(x_i, y_i) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} t_0(x_0, y_0) \cdot h_2(x_i, y_i; x_0, y_0) dx_0 dy_0 = t_0 \otimes h_2$$

$$U(x_i, y_i; x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(x_0, y_0) \cdot t_0(x_s - x_0, y_s - y_0) \cdot h_2(x_i, y_i; x_0, y_0) dx_0 dy_0$$

6.3. 扫描共焦显微镜原理

扫描共焦显微镜的分辨率:

$$U(x_i, y_i; x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(x_0, y_0) \cdot t_0(x_s - x_0, y_s - y_0) \cdot h_2(x_i, y_i; x_0, y_0) dx_0 dy_0$$

$$I(x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(x_i, y_i) \cdot |U(x_i, y_i; x_s, y_s)|^2 dx_i dy_i$$

点物体: $t_0(x_0, y_0) = \delta(x_0, y_0)$ 点探测器: $P(x_i, y_i) = \delta(x_i, y_i)$

$$\begin{aligned} U(0,0; x_s, y_s) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h_1(x_0, y_0) \cdot t_0(x_s - x_0, y_s - y_0) \cdot h_2(x_0, y_0) dx_0 dy_0 \\ &= h_1 h_2 \otimes t_0 \end{aligned}$$

$$I(x_s, y_s) = |h_1 h_2 \otimes t_0|^2$$

扫描共焦显微镜的点扩散函数

若 $h_1 = h_2 = \frac{J_1(t)}{t}$, 则 $I(x_s, y_s) = \left| \frac{J_1(t)}{t} \right|^4$ 其理论分辨率为Rayleigh衍射极限的 $1/\sqrt{2}$ 倍

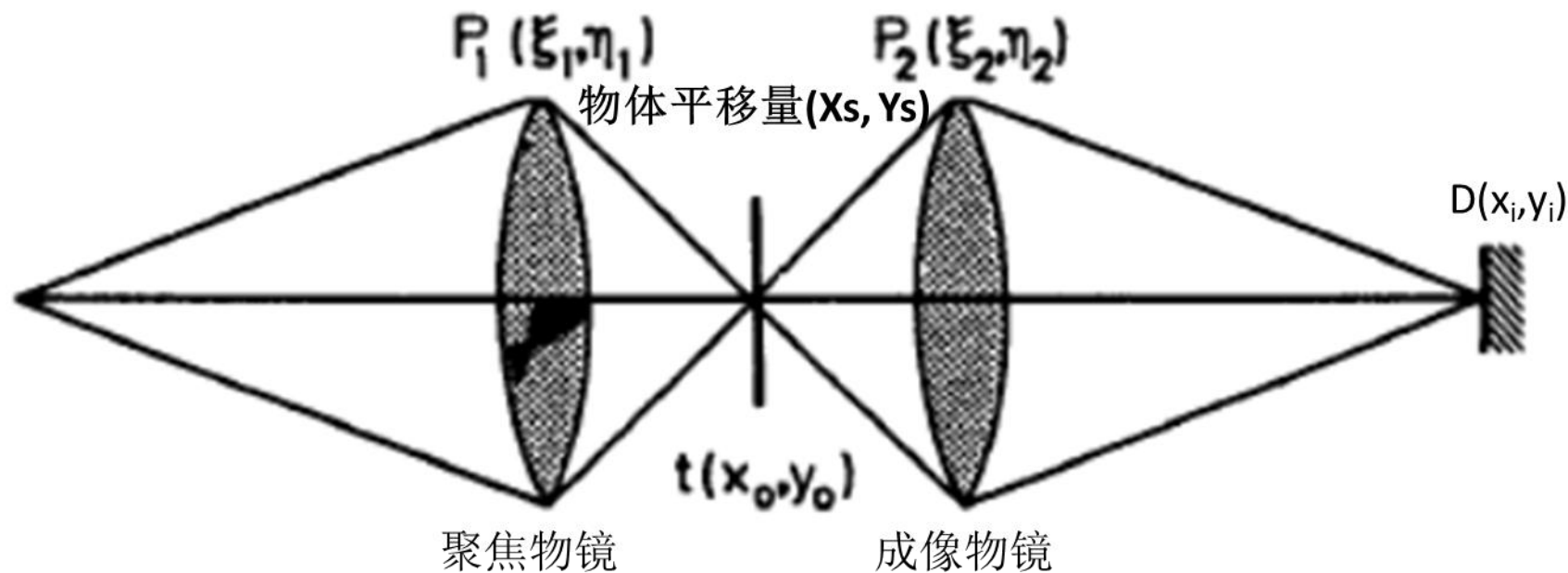
第一个零点: $0.61\lambda/\text{NA}/\sqrt{2}=0.43 \lambda/\text{NA}$ FWHM: $0.515\lambda/\text{NA}/\sqrt{2} = 0.36 \lambda/\text{NA}$

6.3. 扫描共焦显微镜原理

非相干型共焦扫描成像

以上结果均是采用相干光照明，可成为相干型扫描共焦显微镜

若采用非相干光照明，则为非相干型扫描共焦显微镜，其能提高空间分辨率

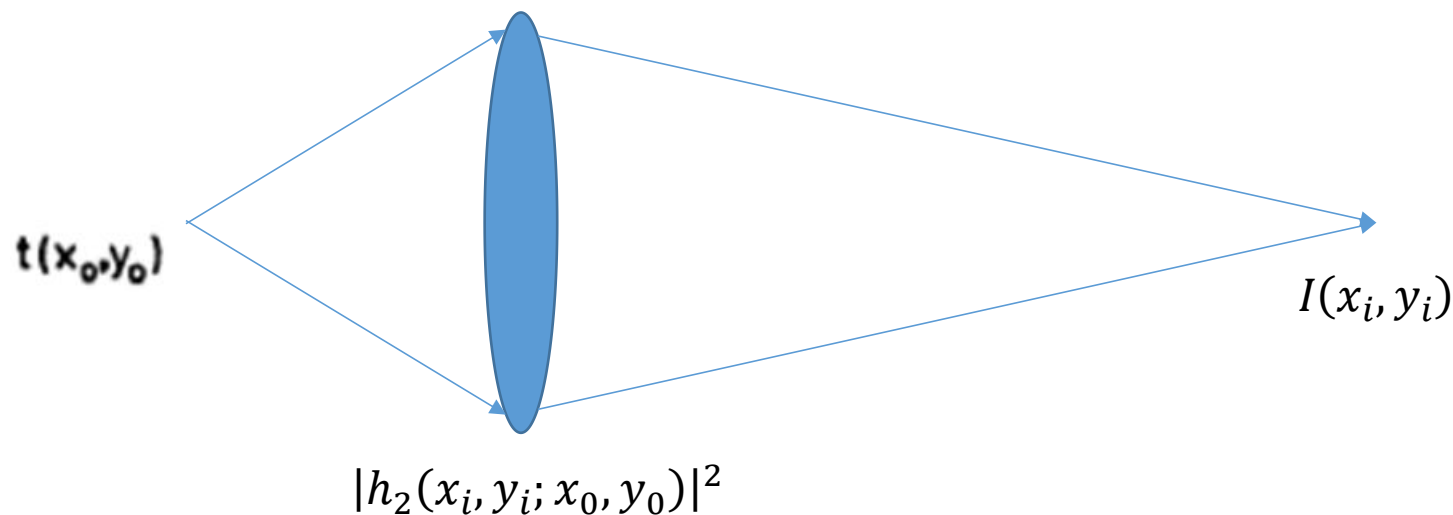


$$I(x_i, y_i; x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |h_1(x_0, y_0)|^2 \cdot t_0(x_s - x_0, y_s - y_0) \cdot |h_2(x_i, y_i; x_0, y_0)|^2 dx_0 dy_0$$

$$I(x_s, y_s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} P(x_i, y_i) \cdot I(x_i, y_i; x_s, y_s) dx_i dy_i$$

6.3. 扫描共焦显微镜原理

非相干型物镜成像：直接成像，无扫描



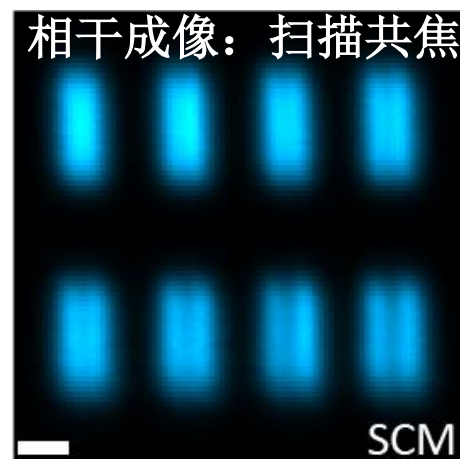
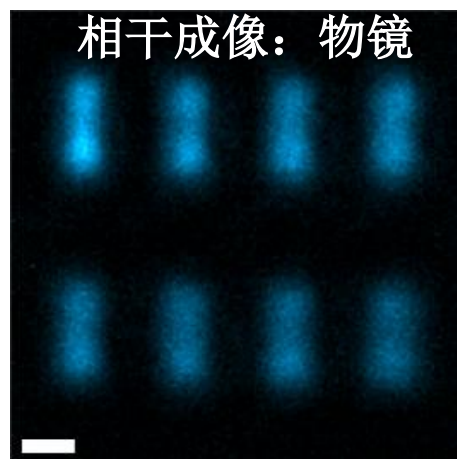
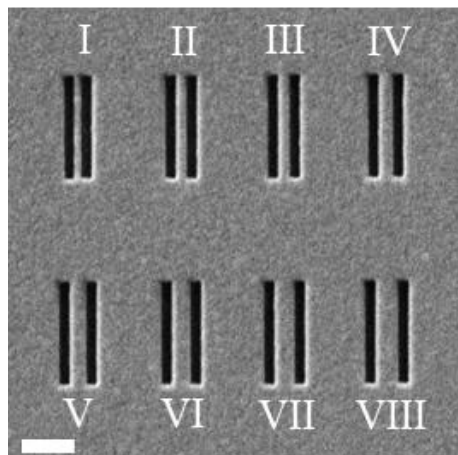
$$U(x_i, y_i) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} t_0(x_0, y_0) \cdot |h_2(x_i, y_i; x_0, y_0)|^2 dx_0 dy_0 = t_0 \otimes |h_2|^2$$

6.3. 扫描共焦显微镜原理

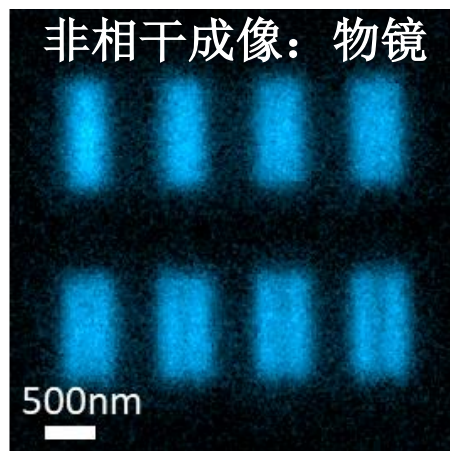
相干和非相干成像对比：

实验

$\lambda=450\text{nm}$, $\text{NA}=0.9$



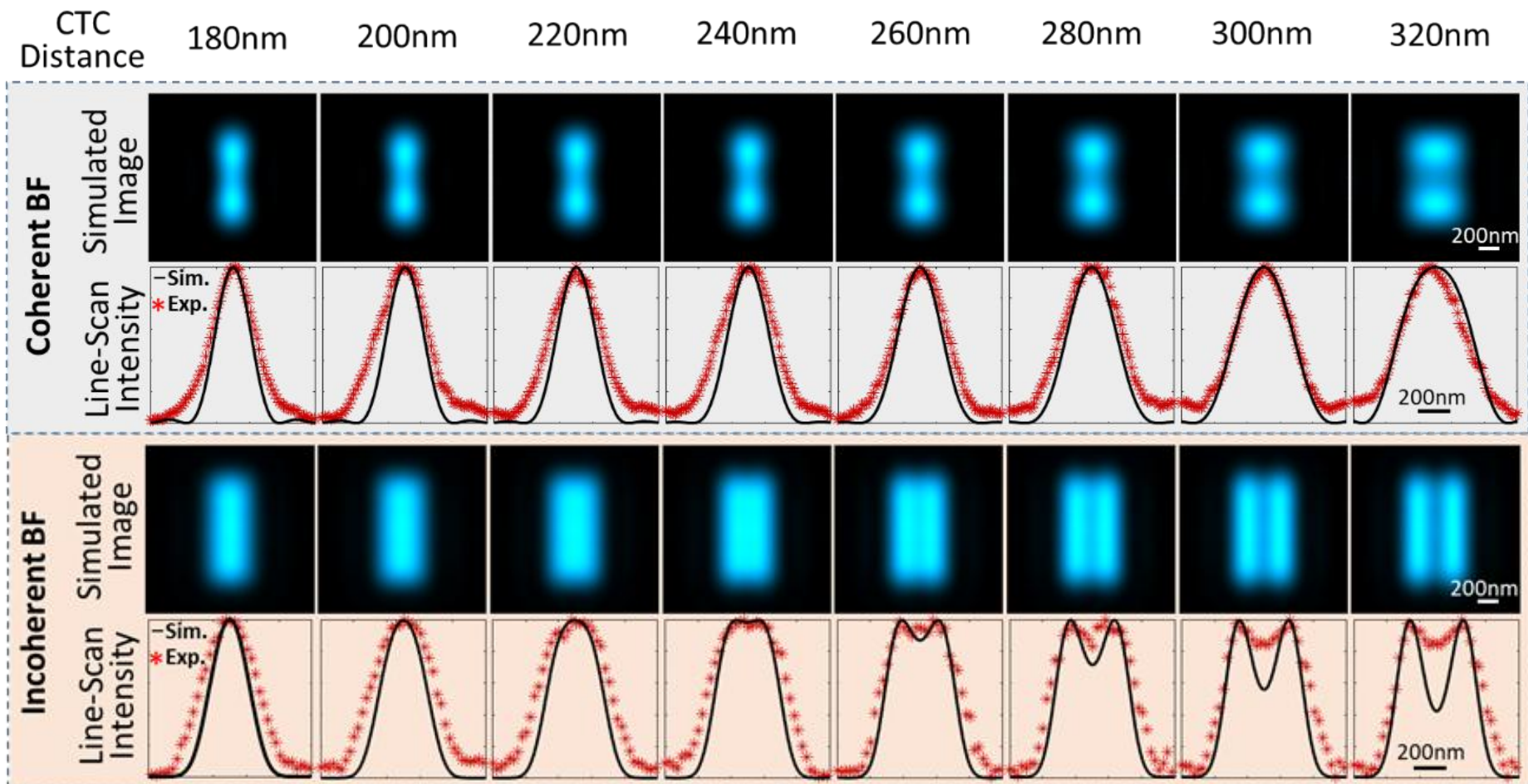
- I: 180nm
- II: 200nm
- III: 220nm
- IV: 240nm
- V: 260nm
- VI: 280nm
- VII: 300nm
- VIII: 320nm



6.3. 扫描共焦显微镜原理

相干和非相干成像对比：

模拟：物镜成像

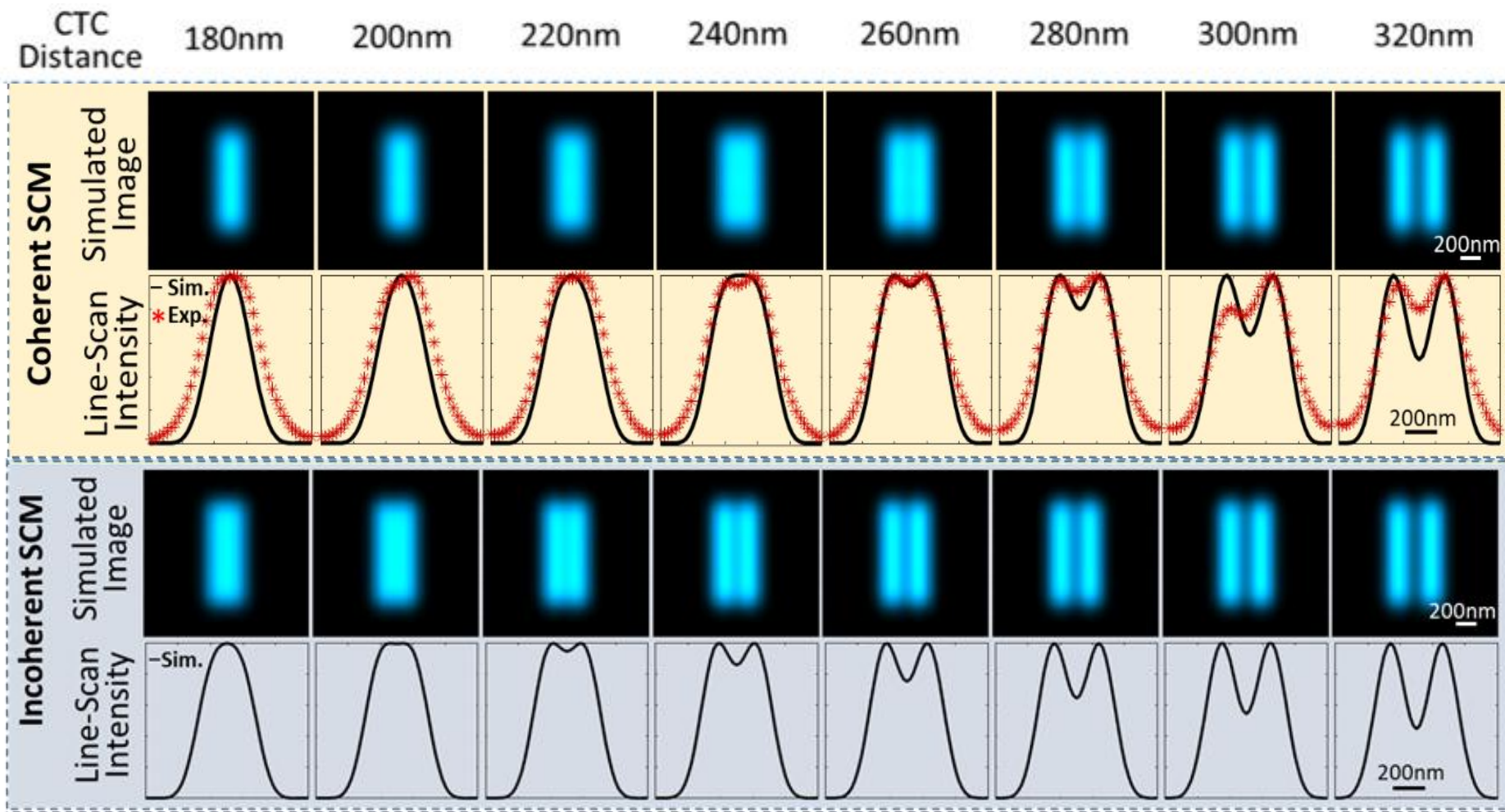


$$0.515 \times 450 / 0.9 = 257\text{nm}$$

6.3. 扫描共焦显微镜原理

相干和非相干成像对比：

模拟：扫描共焦成像



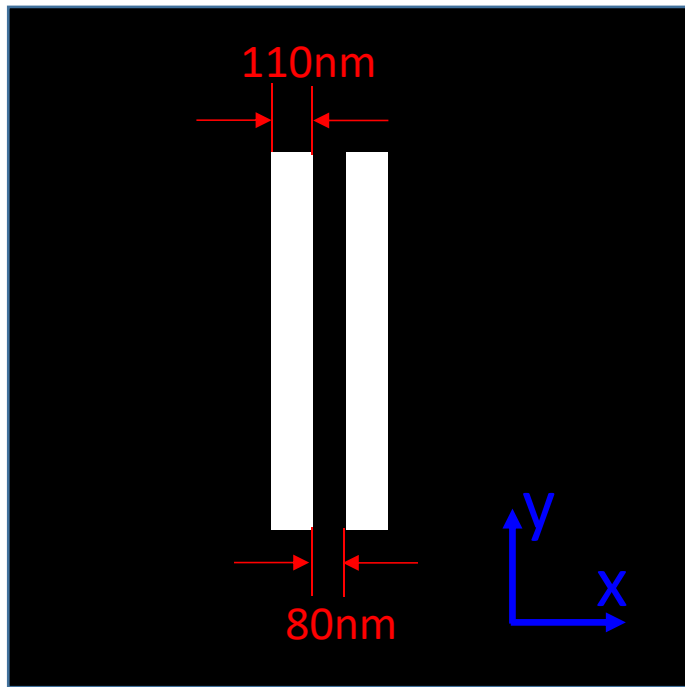
$$0.515 \cdot 450 / 0.9 / \sqrt{2} = 182\text{nm}$$

6.3. 扫描共焦显微镜原理

扫描共焦成像中pinhole尺寸的影响:

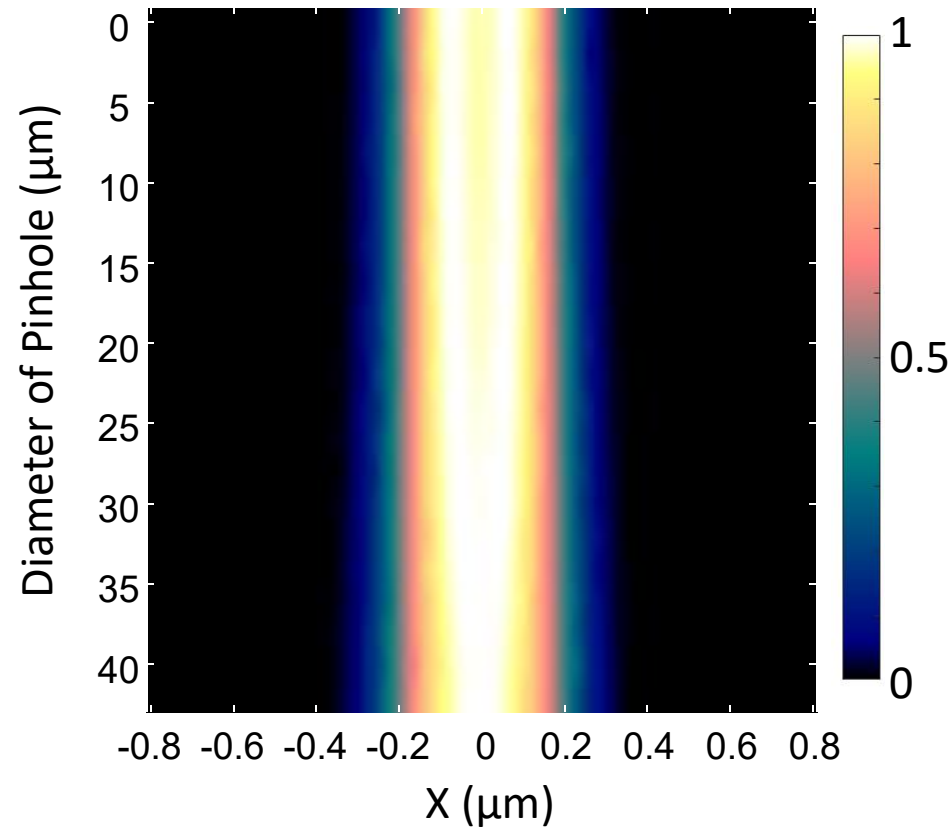
A

Testing Object



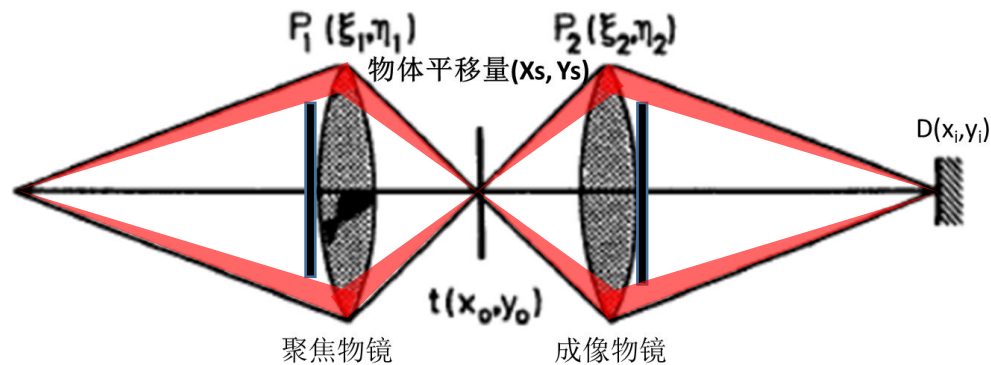
B

Line-scanning Intensity



6.3. 扫描共焦显微镜原理

	相干成像	非相干成像
物镜	Worse than $0.515\lambda/\text{NA}$	$0.515\lambda/\text{NA}$
共焦扫描	$\sim 0.515\lambda/\text{NA}$	$0.515\lambda/\text{NA}/\text{sqrt}(2)$
共焦扫描（环形照明和收集）	$\sim 0.38\lambda/\text{NA}$	$\sim 0.38\lambda/\text{NA}/\text{sqrt}(2)$



6.4. 超分辨成像技术

近场显微镜

1. 近场扫描光学显微镜
2. 近场超透镜(Superlens)
3. 近场微球显微镜

远场显微镜

荧光显微镜

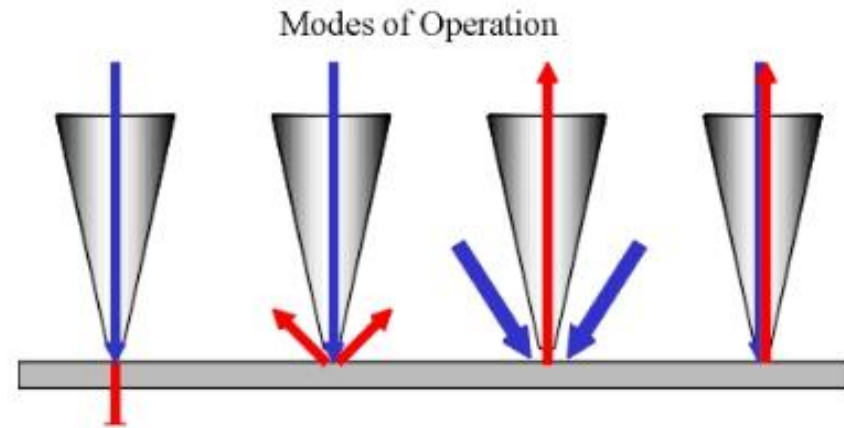
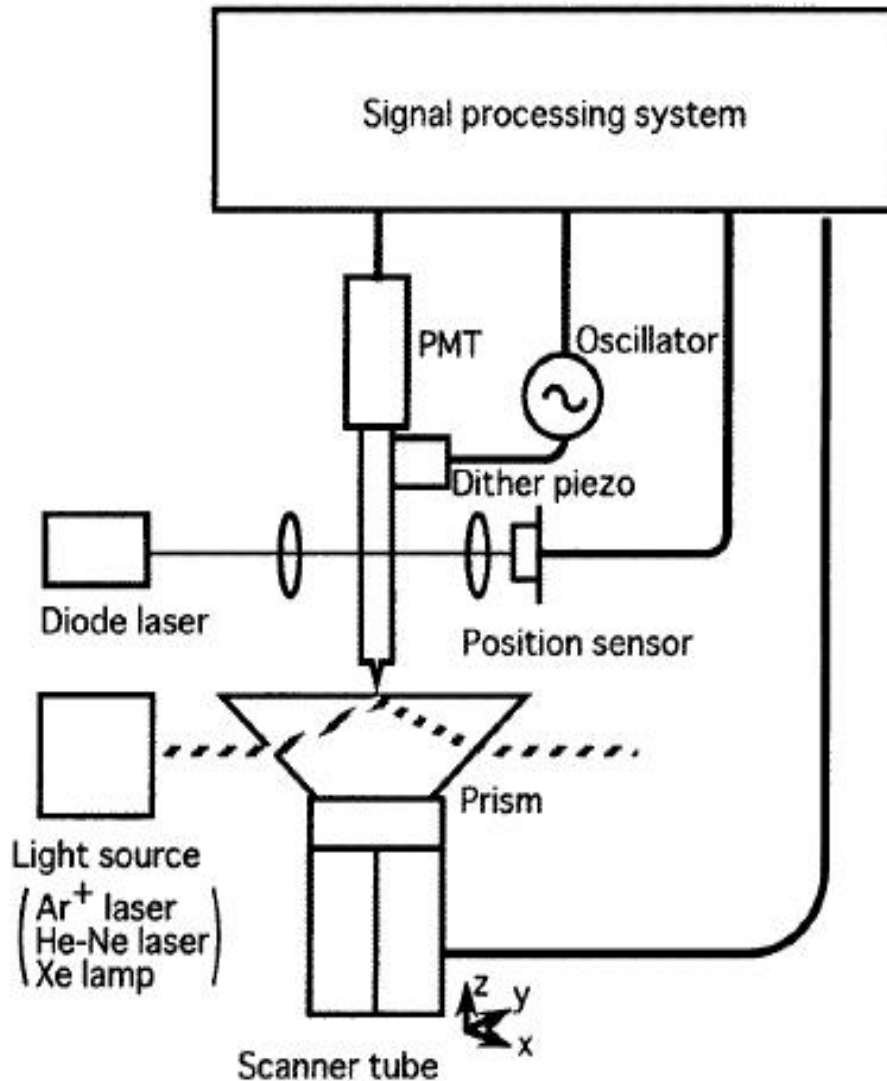
4. STED显微镜
5. PLAM/STORM
6. MINI-FLUX
7. 结构光显微镜

非荧光显微镜

8. 基于超震荡\超临界透镜的共焦扫描显微镜
9. 压缩感知的相干衍射成像

6.4. 超分辨成像技术

1. 近场扫描光学显微镜

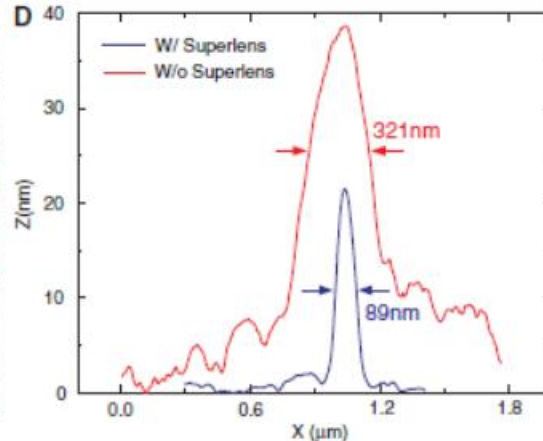
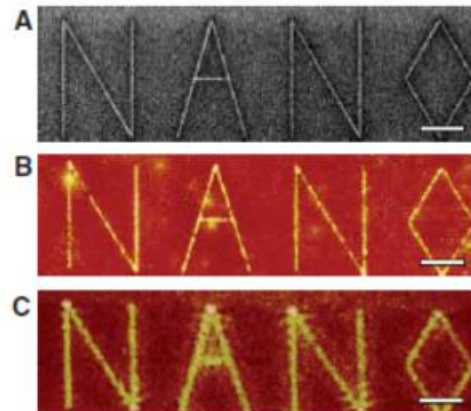
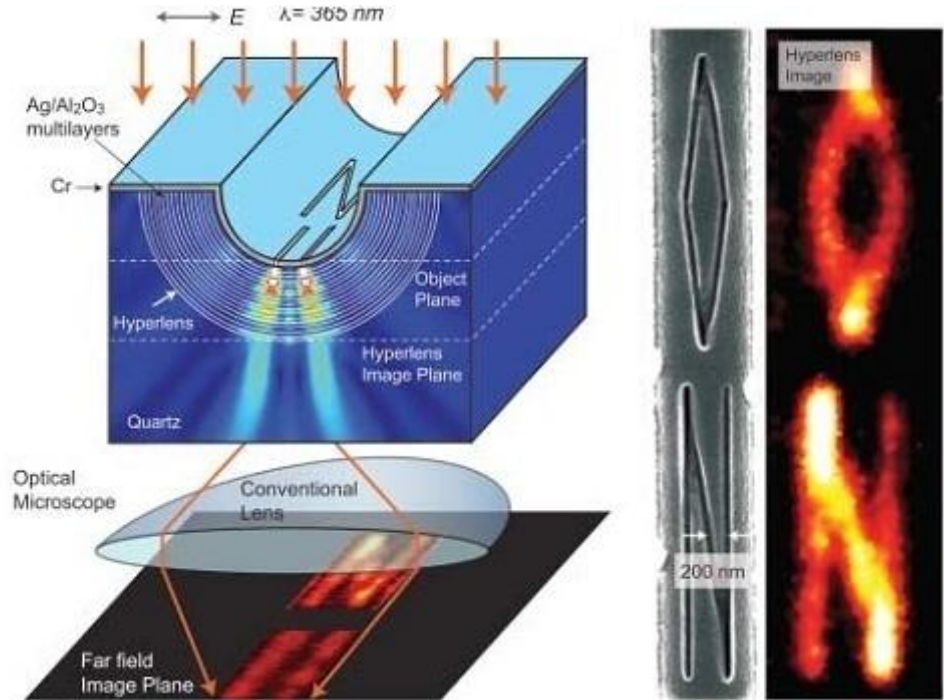
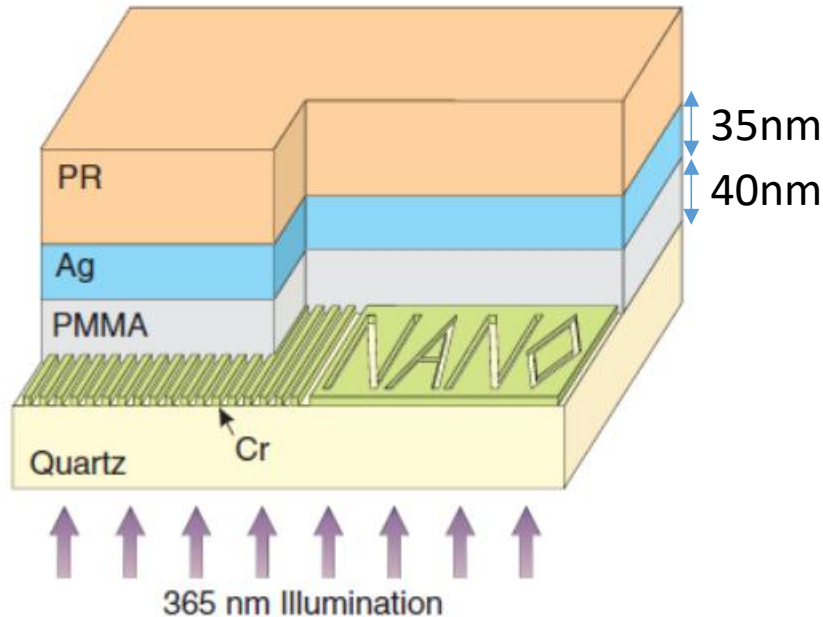


优点：分辨率高， $<10\text{nm}$

缺点：工作距离短、易损伤样品、需经常换Tip、操作较困难

6.4. 超分辨成像技术

2. 近场超透镜(Superlens)

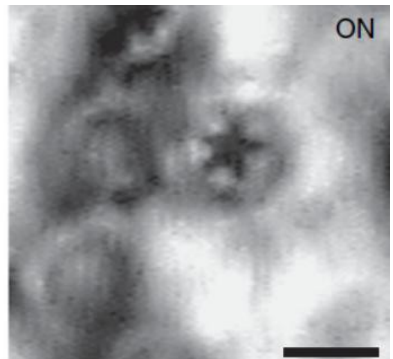
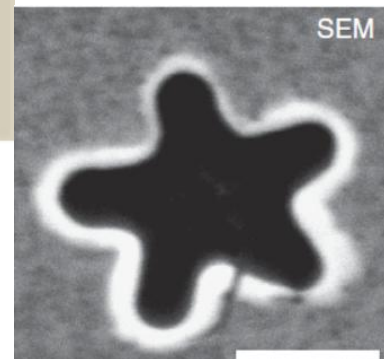
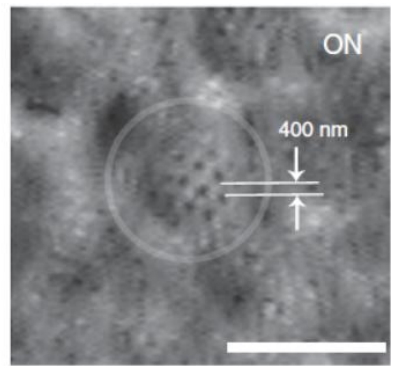
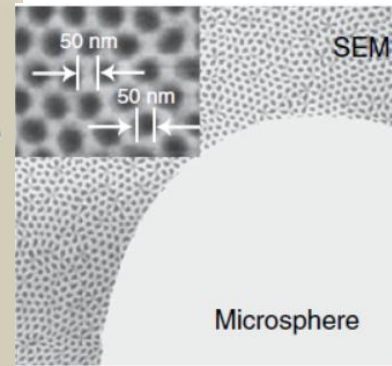
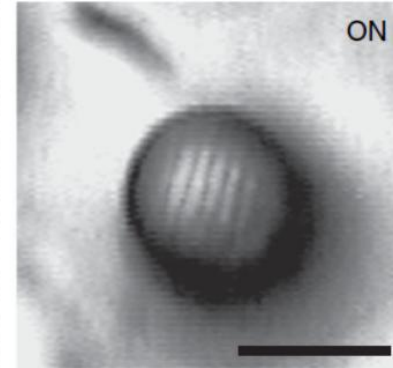
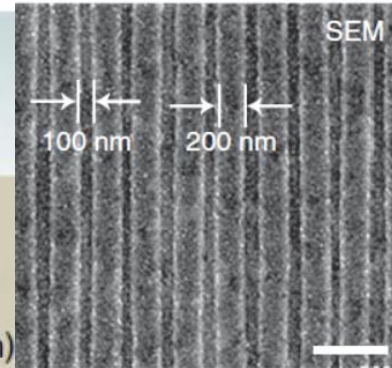
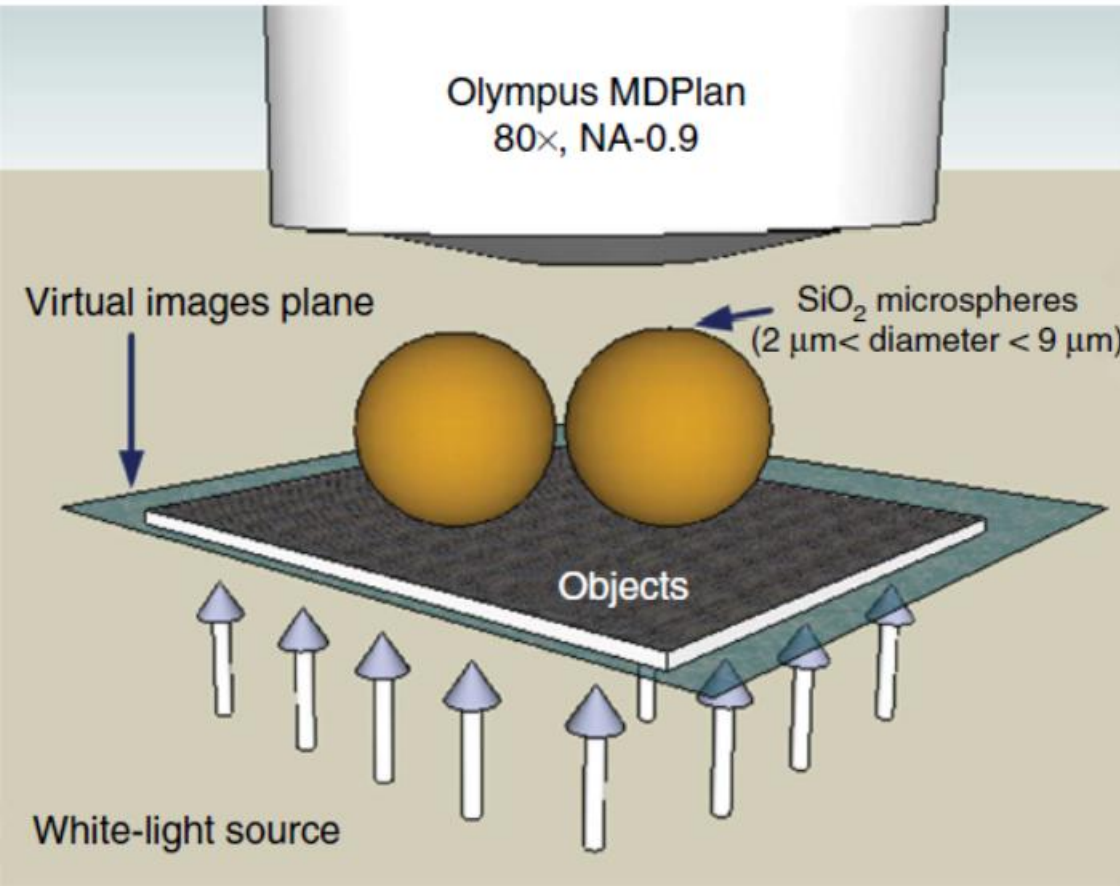


优点：分辨率高， $<60 \text{ nm}$

缺点：工作距离短、易损伤样品、操作较困难、物体必须紧贴透镜

6.4. 超分辨成像技术

3. 近场微球显微镜



优点：分辨率高，<25nm ?

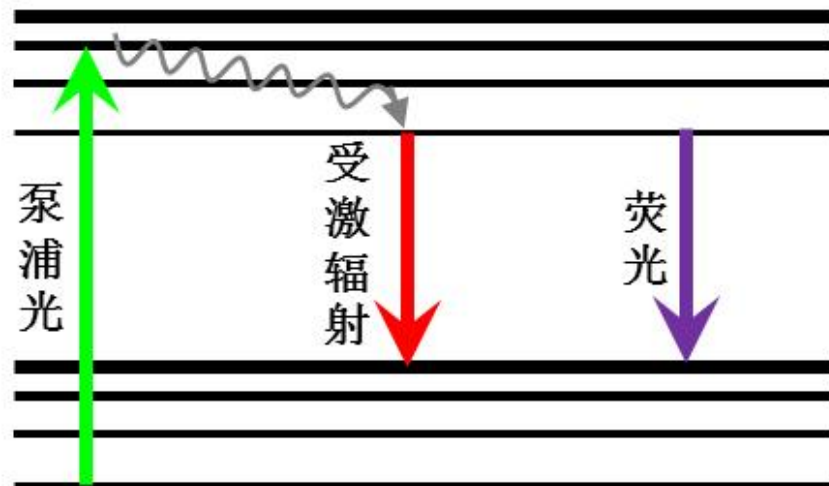
缺点：工作距离短、易损伤样品、操作较困难、物体必须紧贴透镜

6.4. 超分辨成像技术

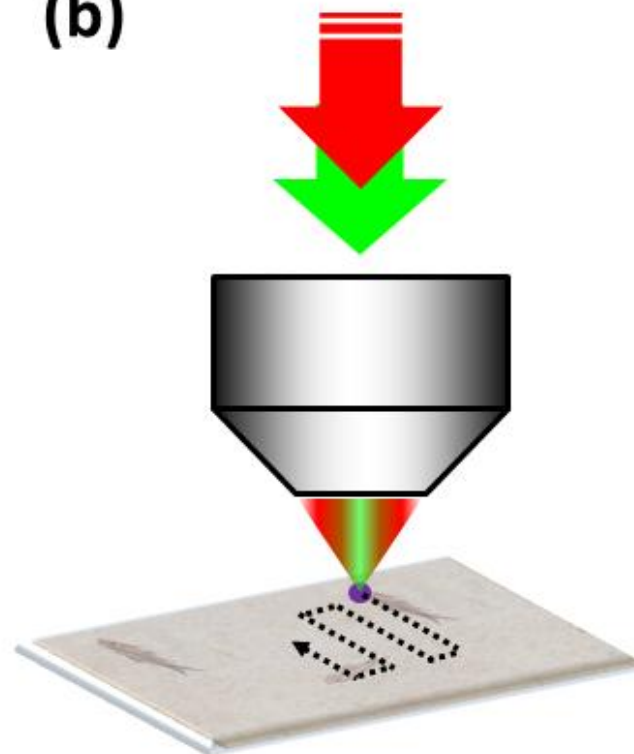
4. STED显微镜

STED扫描成像过程

(a)



(b)



1) 绿色的激发光束可以把荧光分子从“关闭”状态切换成明亮的“发光”状态

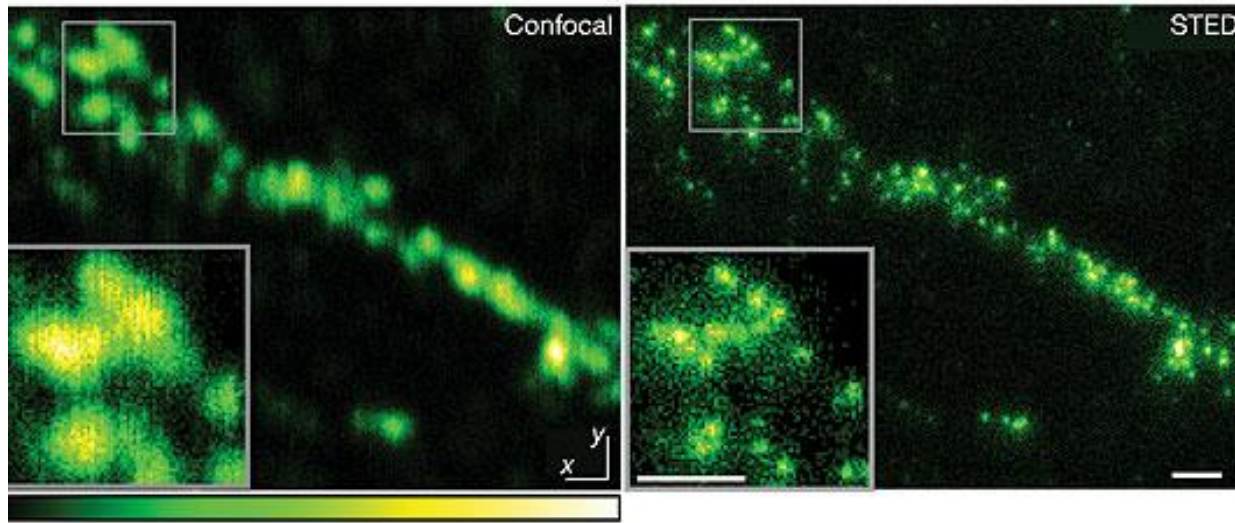
3) 只记录中央分子的发光，其发光面积取决于环形光束的内径

2) 环形光束把除中央分子之外的分子“关闭”、让其不再发光。

$$\Delta r \approx \frac{\lambda}{2n \sin \alpha \sqrt{1 + I_{\max}/I_s}}$$

6.4. 超分辨成像技术

4. STED显微镜



共聚焦显微图像

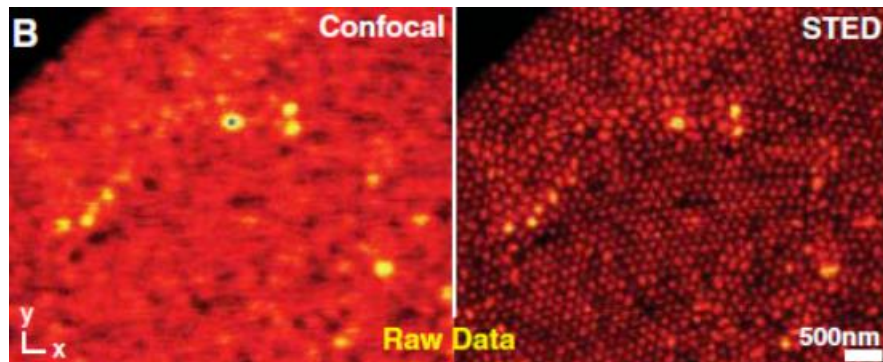
无法区分单个突触囊泡

(直径大约40 nm)

STED显微图像

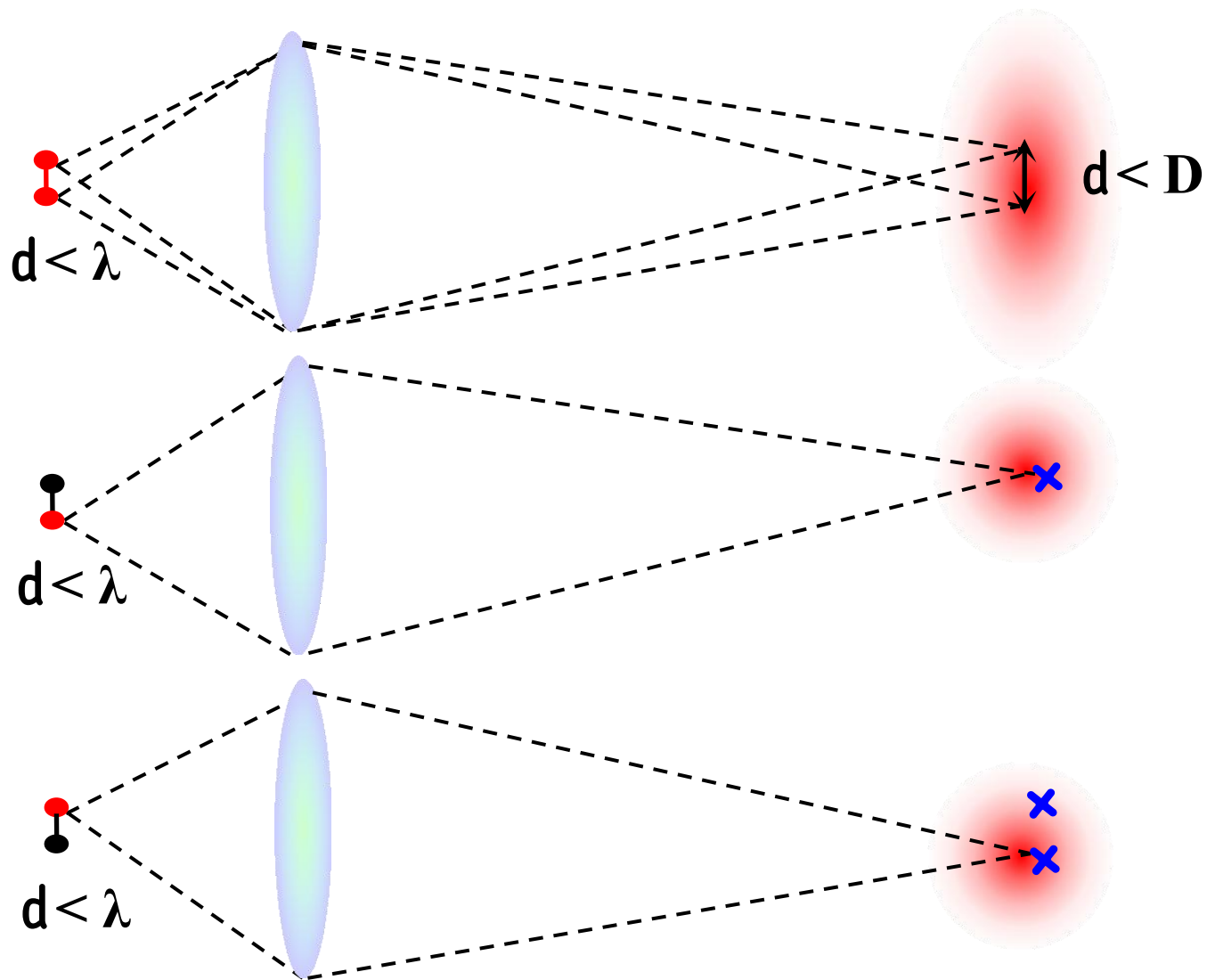
观察到单个突触囊泡

Nature, 440, 935(2006)



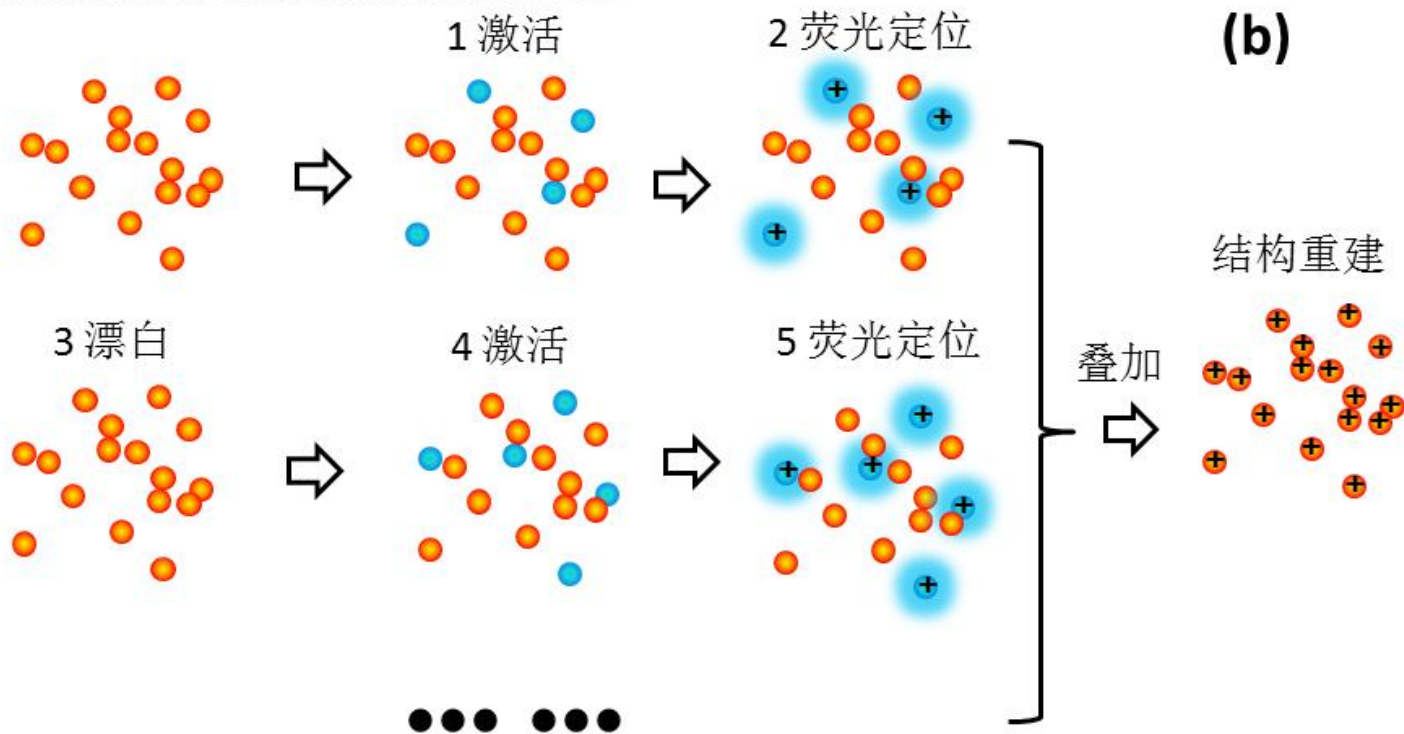
6.4. 超分辨率成像技术

5. PLAM/STORM显微镜



6.4. 超分辨率成像技术

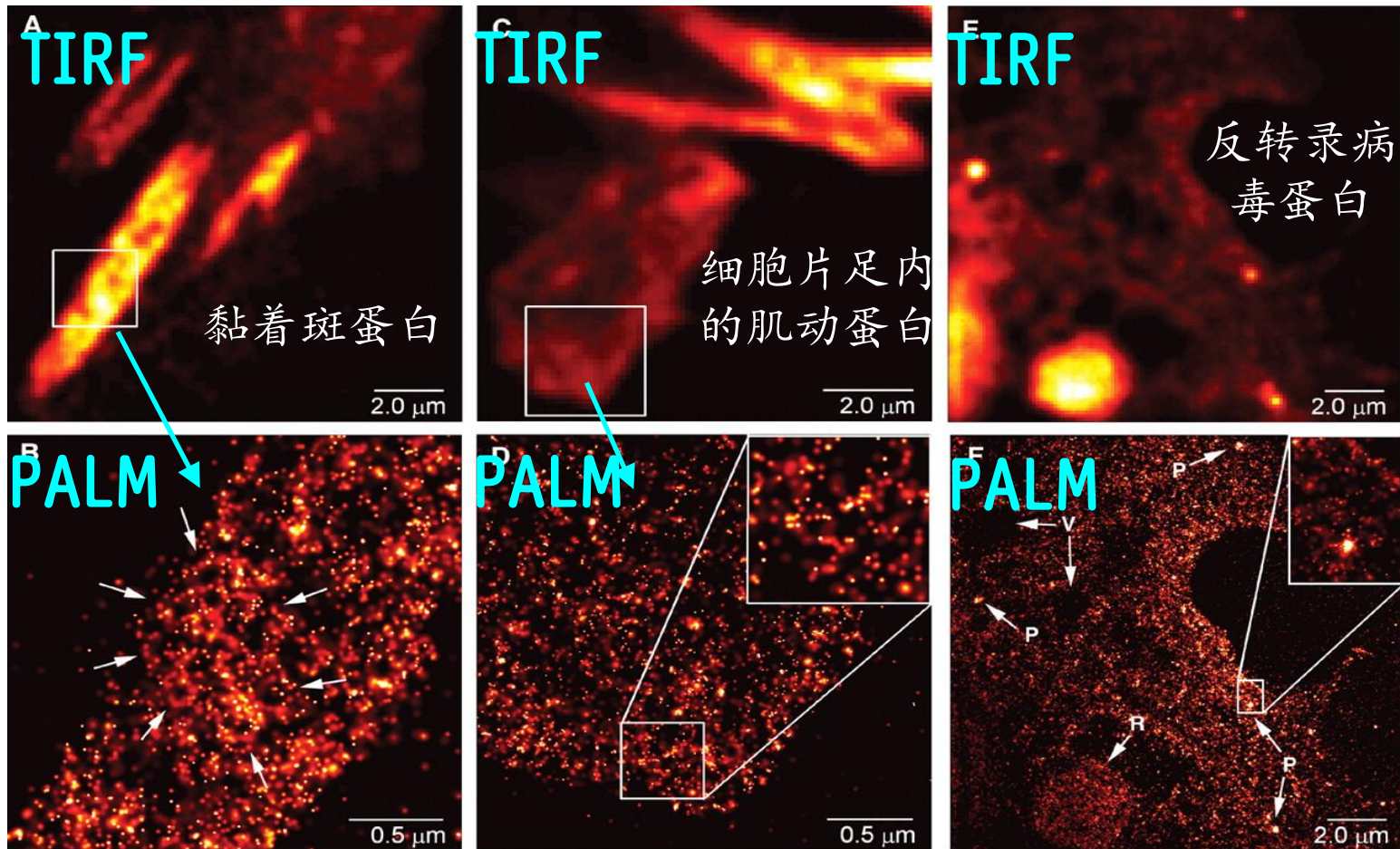
5. PLAM/STORM显微镜



6.4. 超分辨成像技术

5. PLAM/STORM显微镜

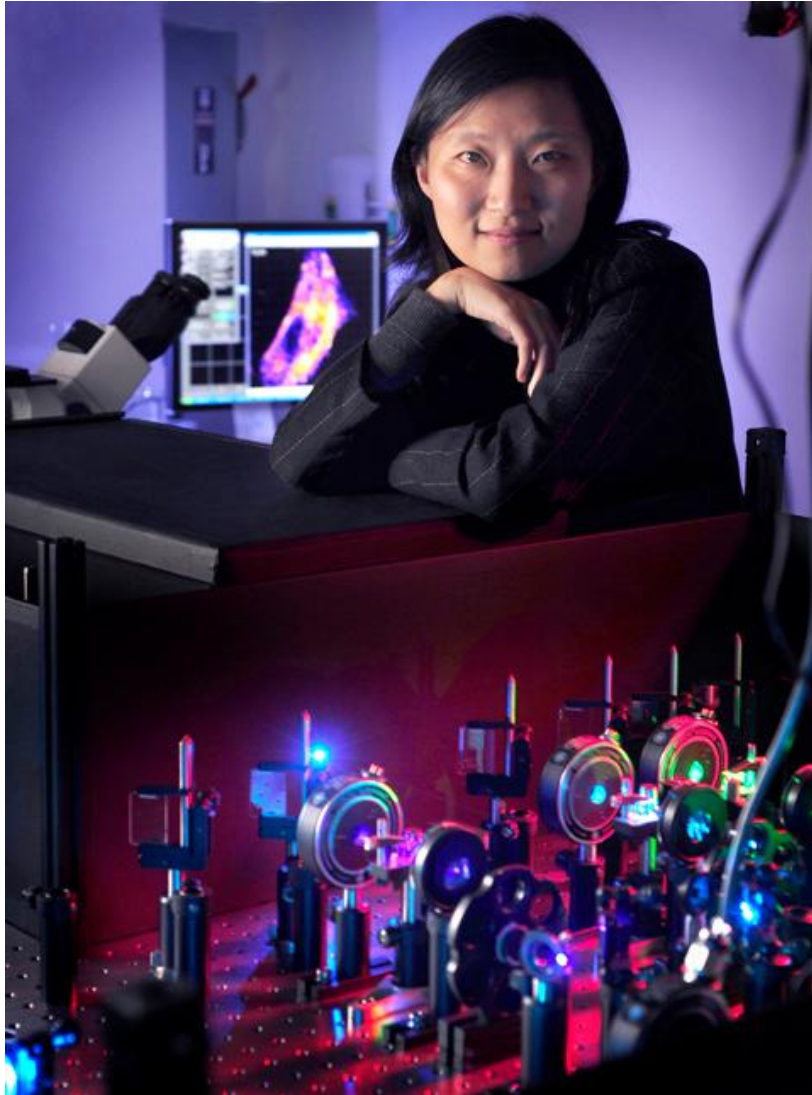
利用PALM技术，将密布于仅为头发丝截面的1/5000的面积中的十万个分子进行高分辨观测，分子间隔约为20纳米。



Science 313,1642 (2006)

6.4. 超分辨成像技术

5. PLAM/STORM显微镜 随机光学重建显微技术(STORM)



PALM 和 STORM 的区别:

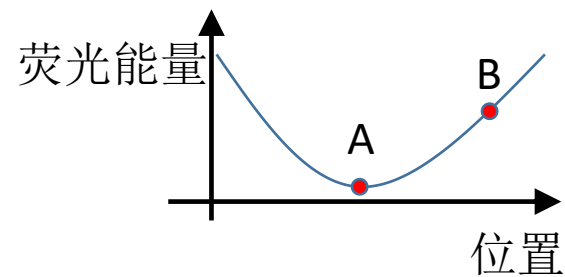
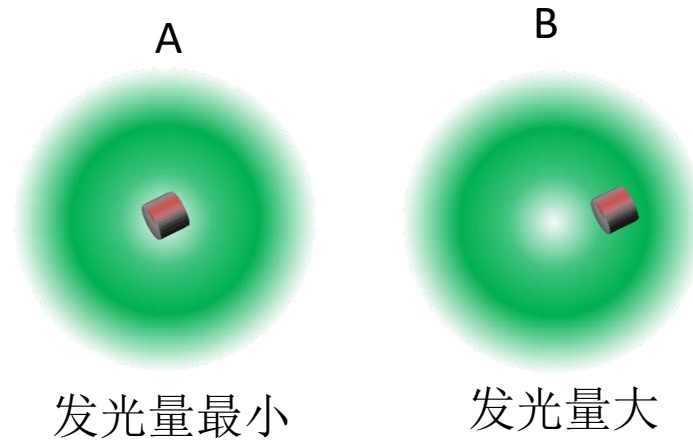
原理一致，标记方法不同

PALM使用从外部引入的光激活蛋白基因（由2009年诺贝尔奖获得者钱永健教授和本次诺贝尔奖获得者莫尔纳尔发现的光激活绿色荧光蛋白分子）标记样品；

STORM使用化学合成的有机荧光分子对（Cy5-Cy3）进行细胞内生物蛋白标记。

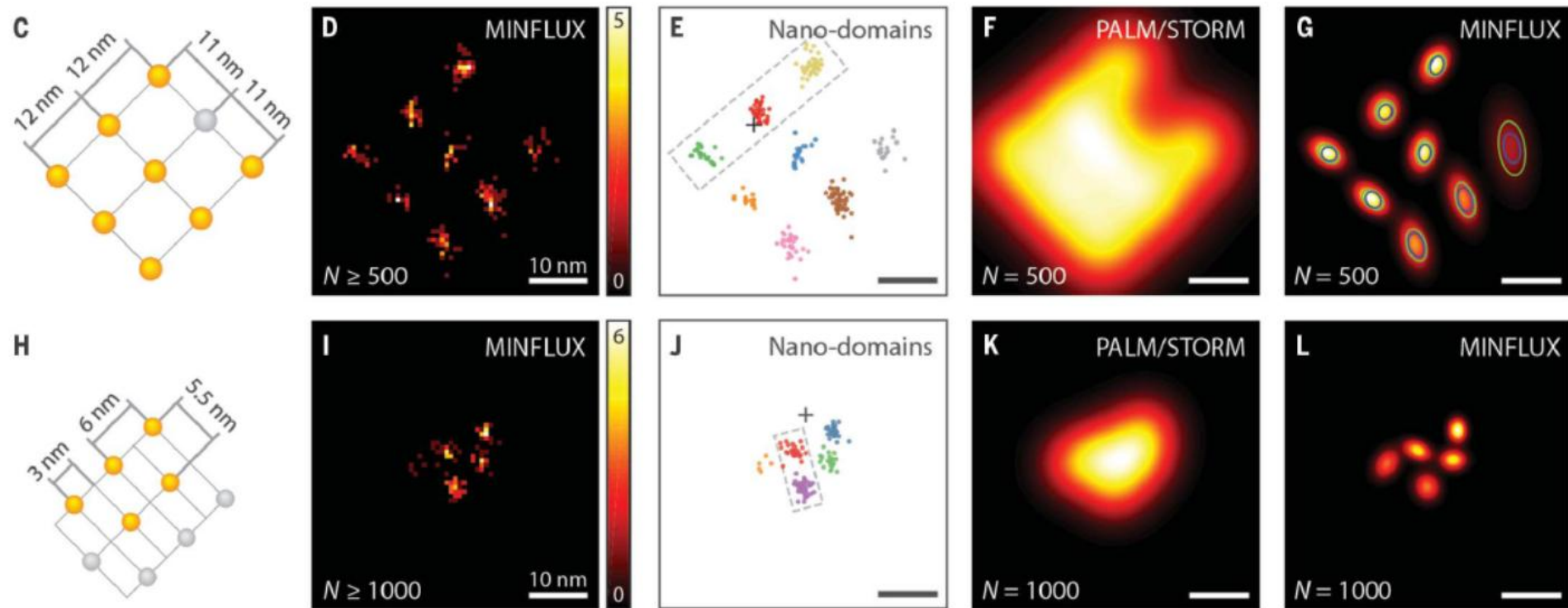
6.4. 超分辨率成像技术

6. MINI-FLUX显微镜



6.4. 超分辨成像技术

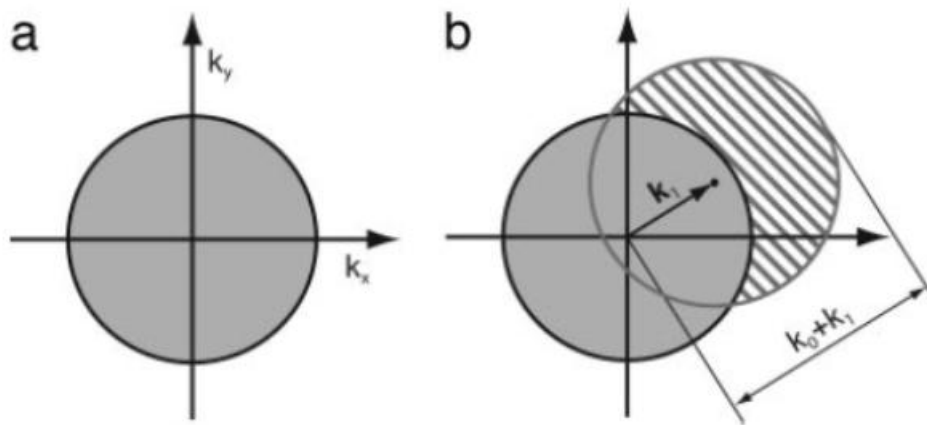
6. MINI-FLUX显微镜



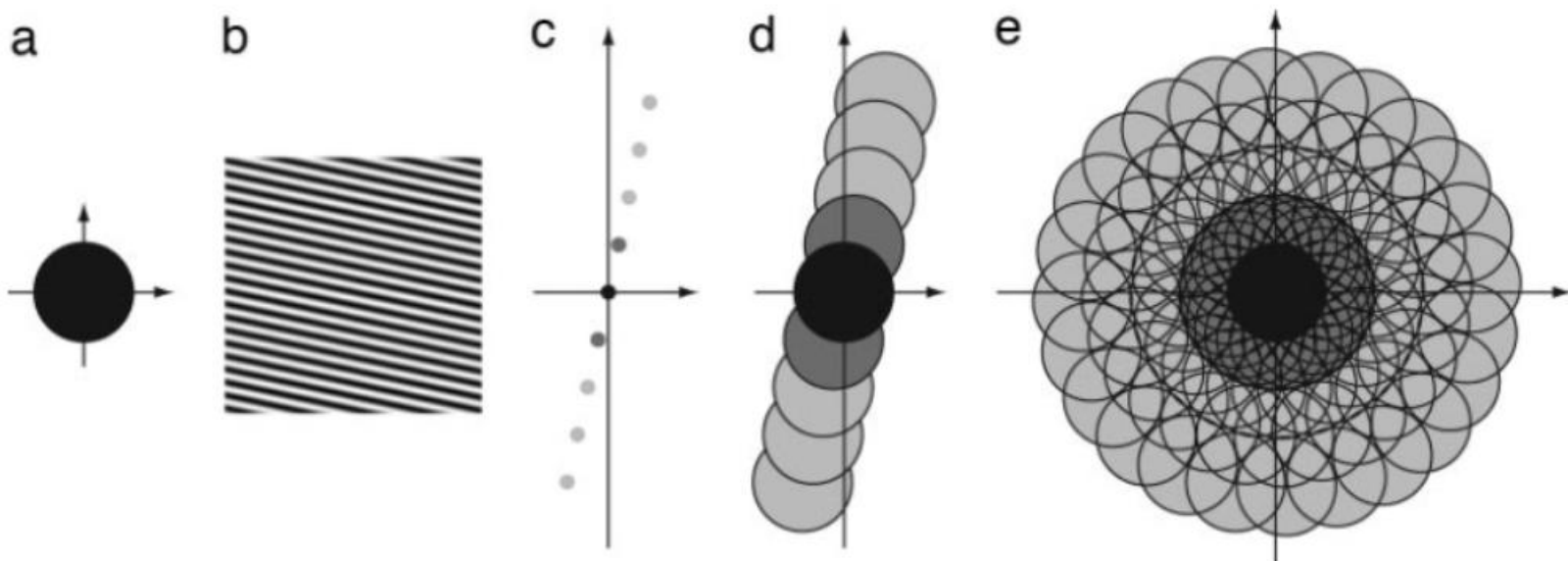
6.4. 超分辨成像技术

7. 结构荧光显微镜

线性光照明

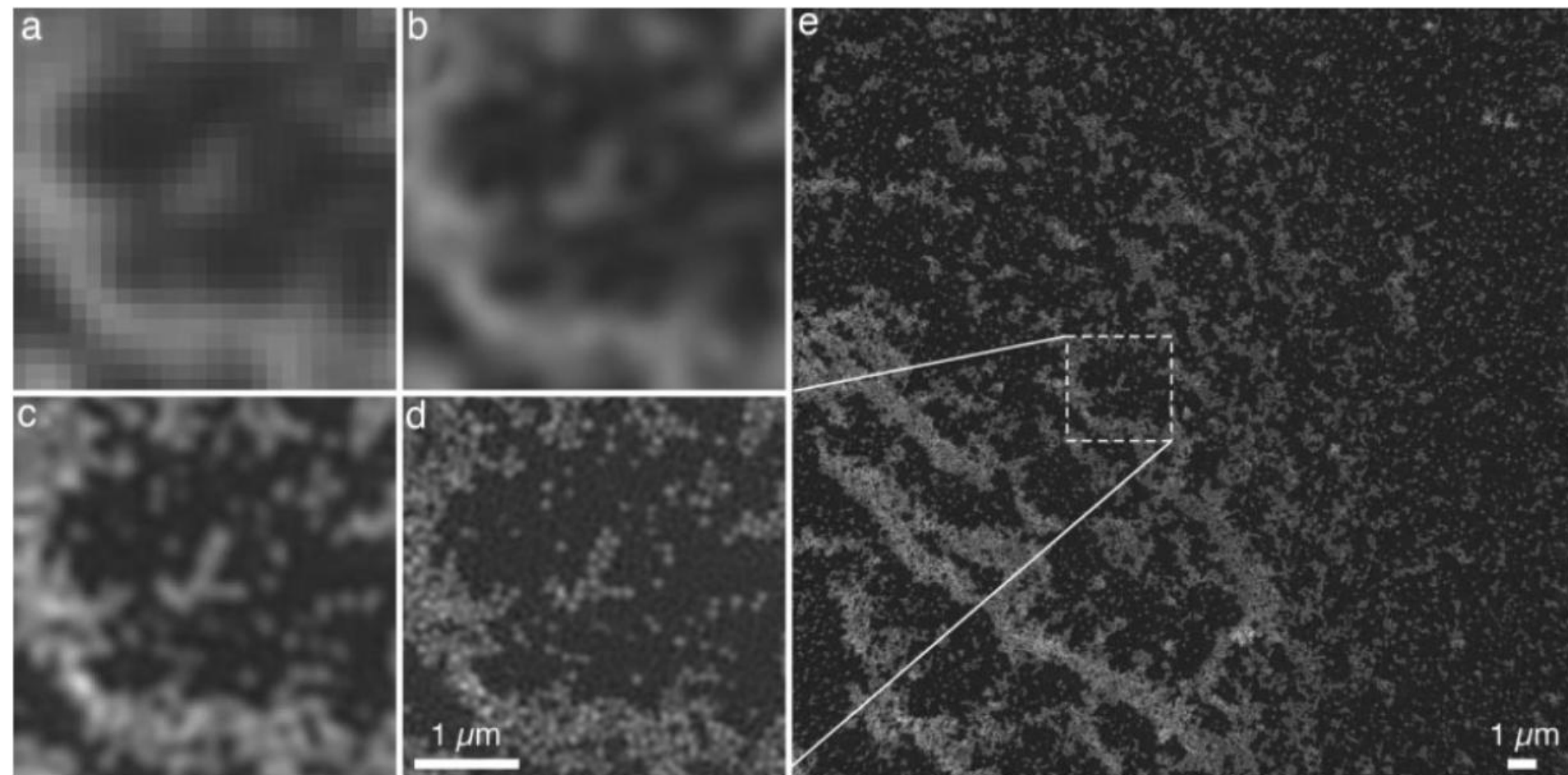


非线性光照明



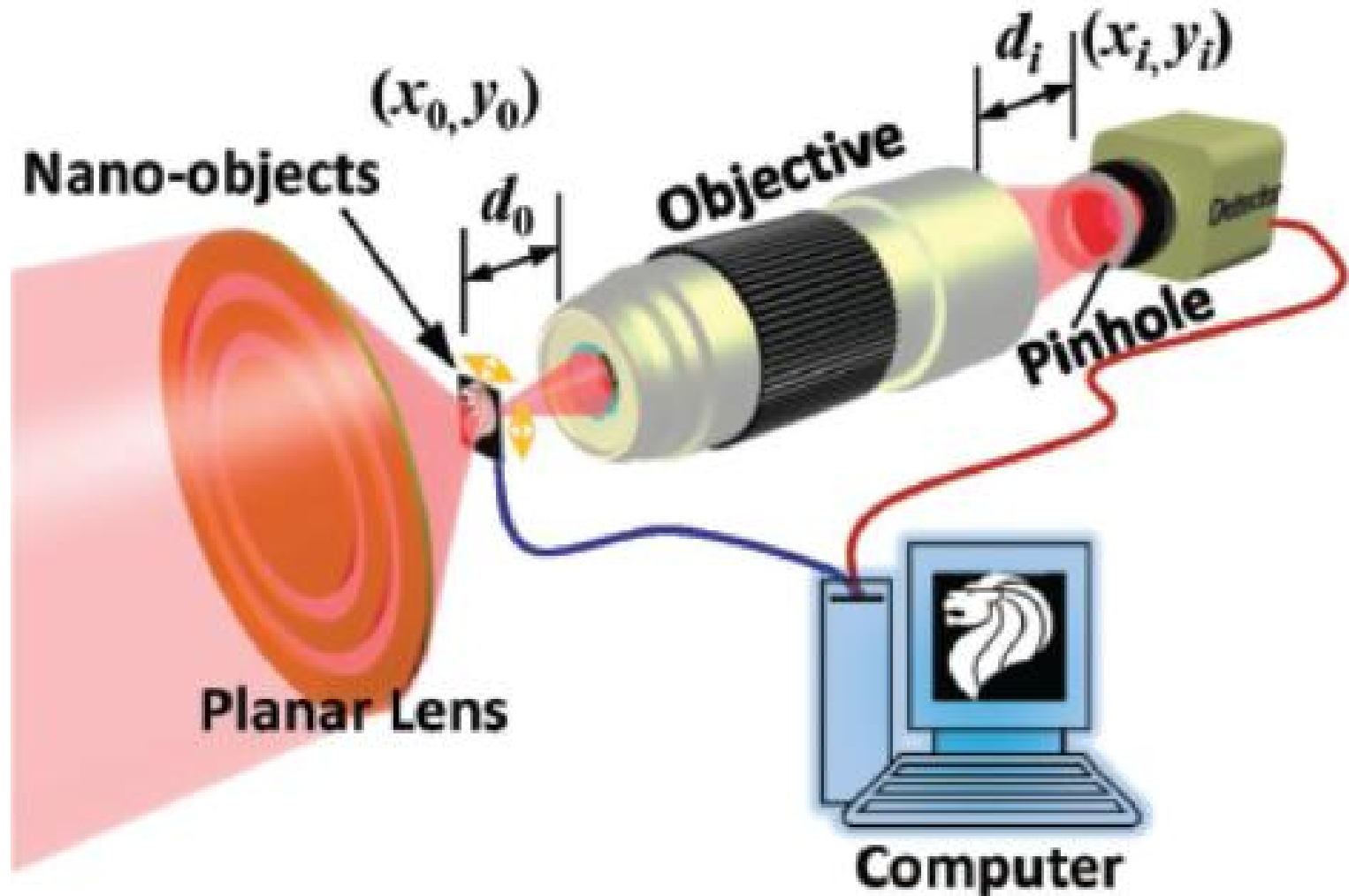
6.4. 超分辨成像技术

7. 结构荧光显微镜



6.4. 超分辨成像技术

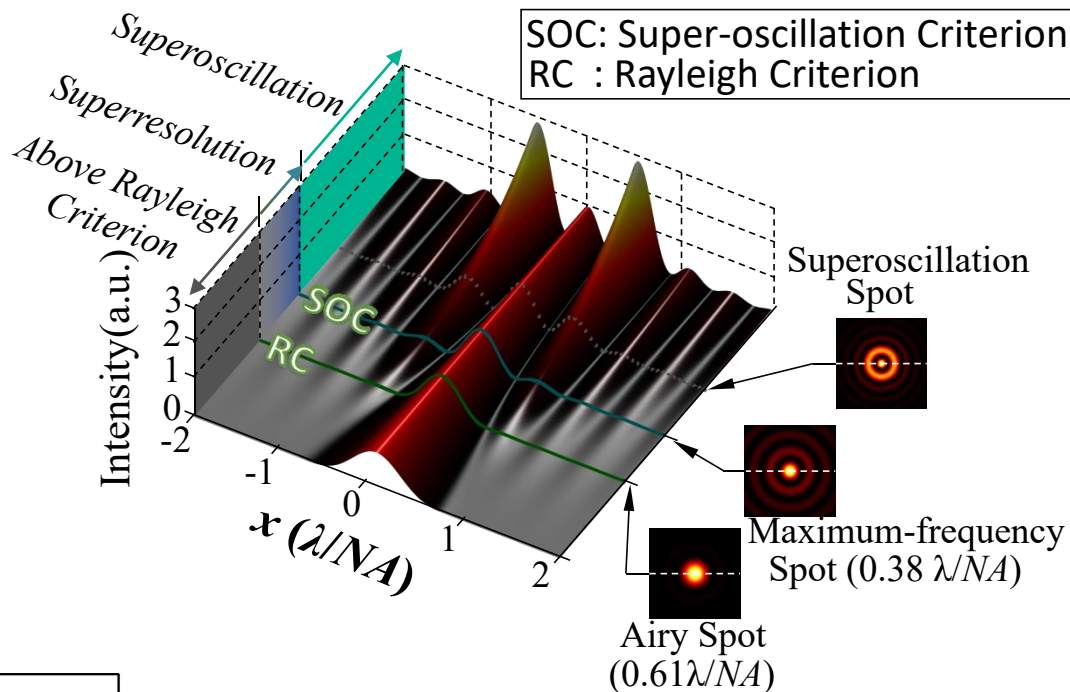
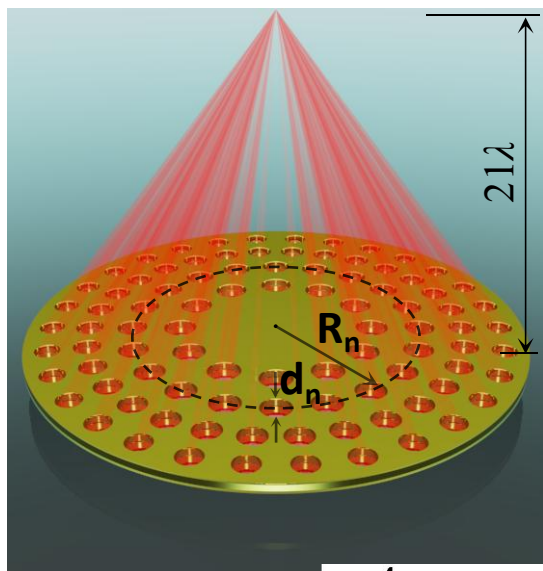
8. 基于超震荡\超临界透镜的共焦扫描显微镜



6.4. 超分辨成像技术

8. 基于超震荡\超临界透镜的共焦扫描显微镜

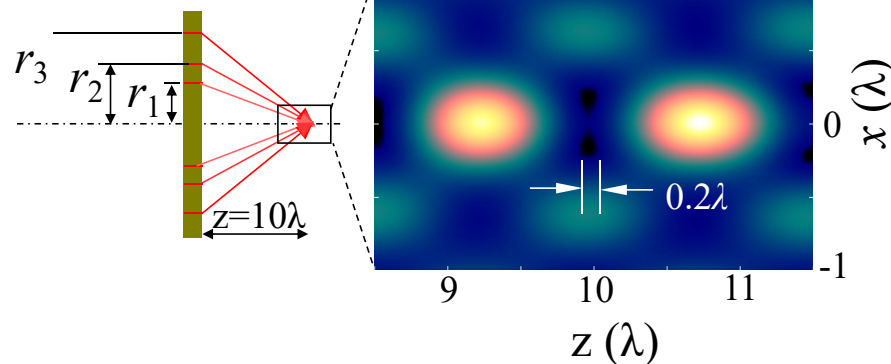
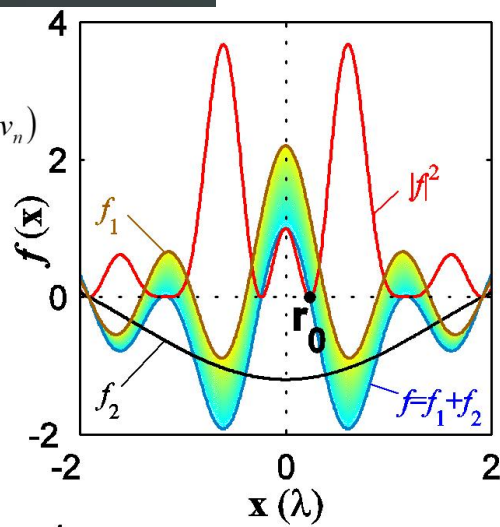
超震荡



$$f(x) = \sum_{n=1}^2 f_n = \sum_{n=1}^2 C_n J_0(k_0 x v_n)$$

$$f_1 = 2.041 J_0(0.98 k_0 r)$$

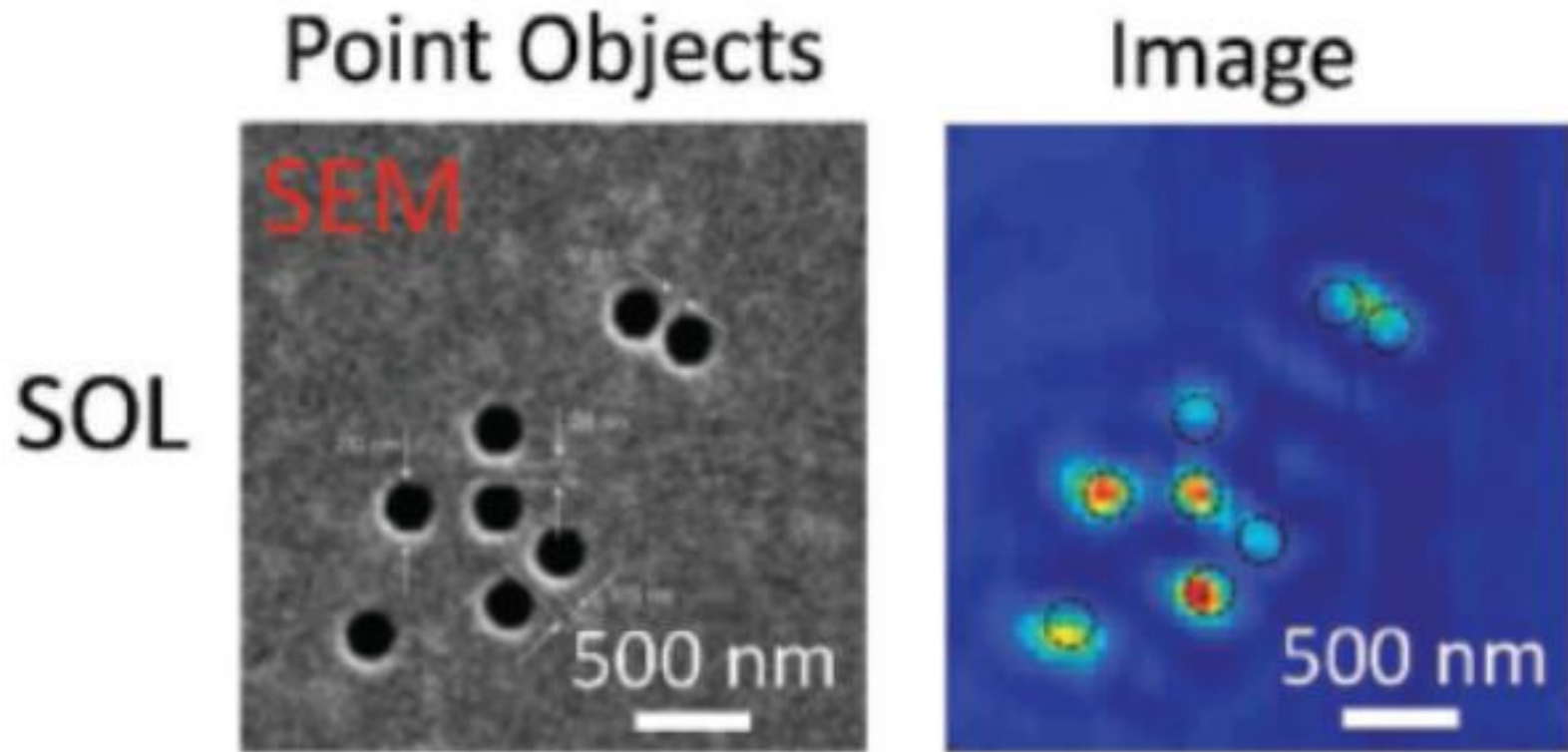
$$f_2 = -1.041 J_0(0.2 k_0 r)$$



6.4. 超分辨成像技术

8. 基于超震荡\超临界透镜的共焦扫描显微镜

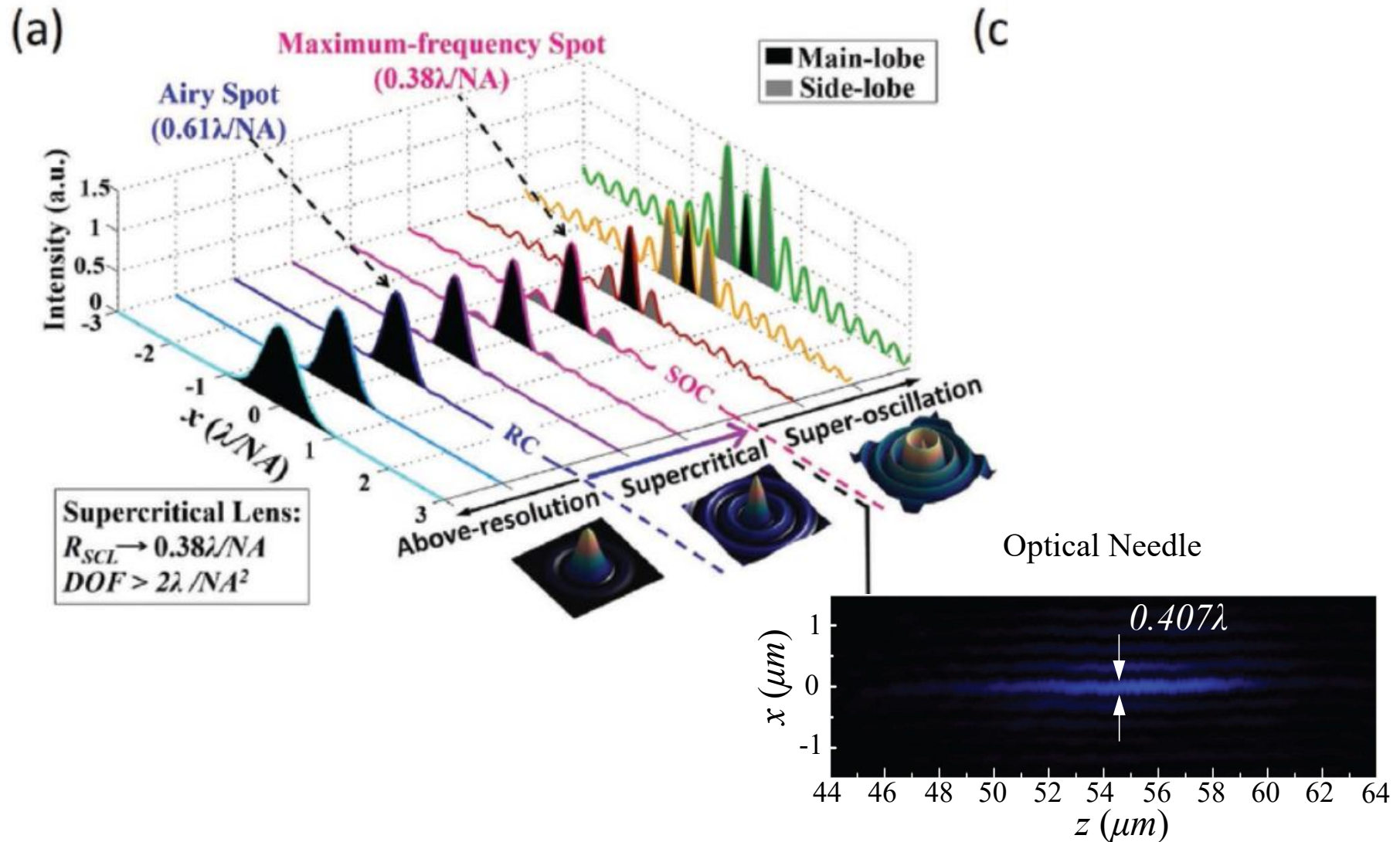
超震荡型SCM



6.4. 超分辨率成像技术

8.基于超震荡\超临界透镜的共焦扫描显微镜

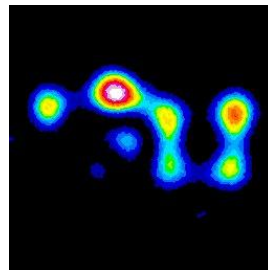
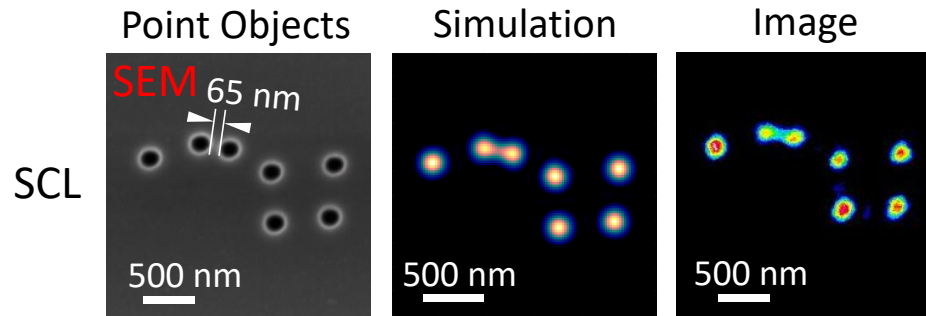
超临界



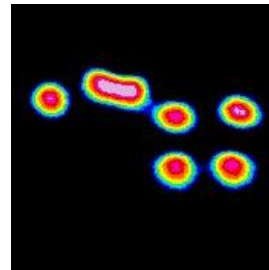
6.4. 超分辨成像技术

8. 基于超震荡\超临界透镜的共焦扫描显微镜

超临界型SCM



Transmission
Microscopy



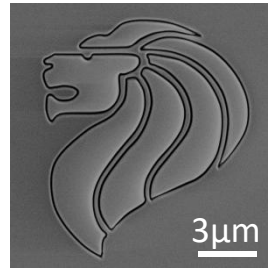
Scanning
Confocal
Microscopy

6.4. 超分辨成像技术

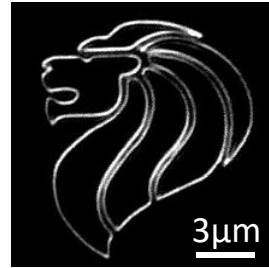
8. 基于超震荡\超临界透镜的共焦扫描显微镜

超临界型SCM

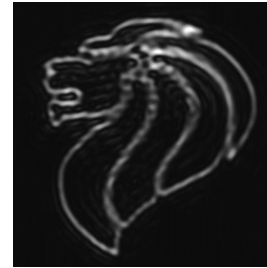
Complex Objects



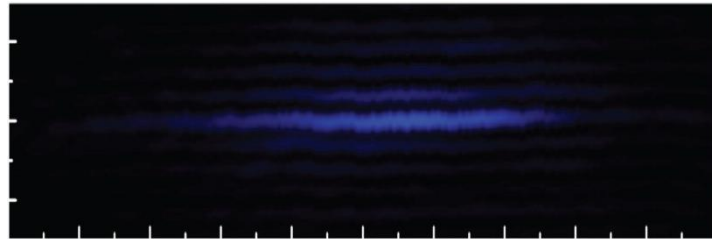
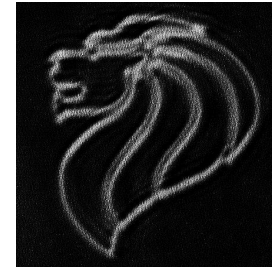
Image



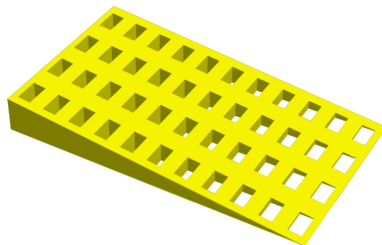
Transmission Microscopy



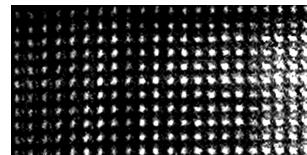
Scanning Confocal Microscopy



3-dimensional Objects

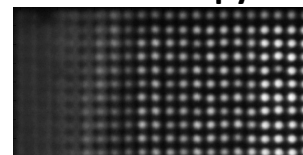


Image



3μm

Transmission Microscopy



Scanning Confocal Microscopy



High-resolution Out-of-Focus Imaging

6.4. 超分辨率成像技术

9. 压缩感知的相干衍射成像

